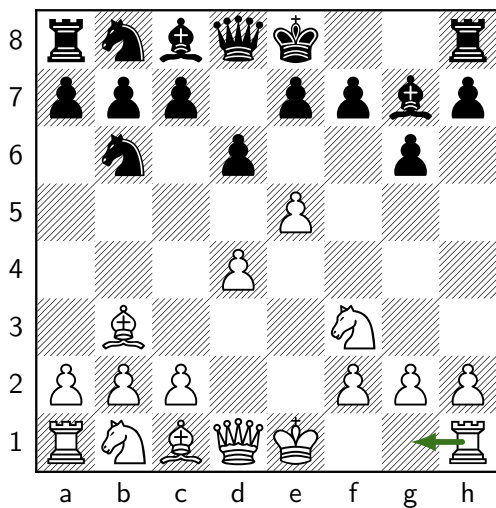


Melanie-ANALOGY — Alekhine Defense

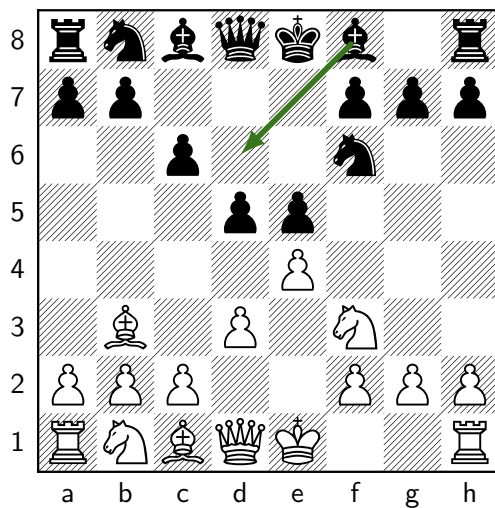
Melanie (Deep_Vision)

Abstract

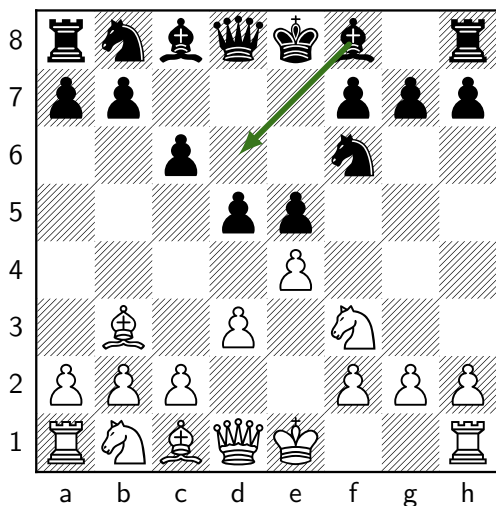
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie’s move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen’s role) are called out, as no material/eval engine explains a move that way.



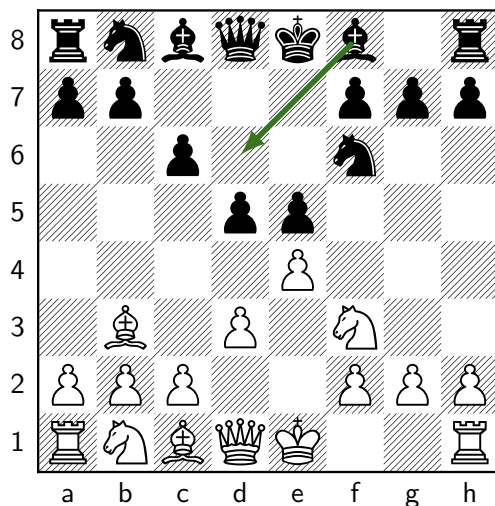
(a)



(b)

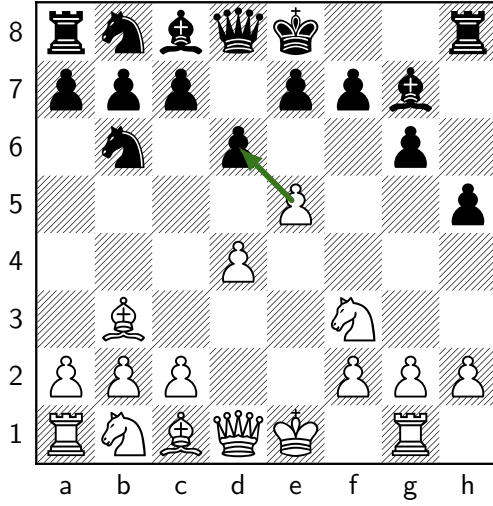


(c)

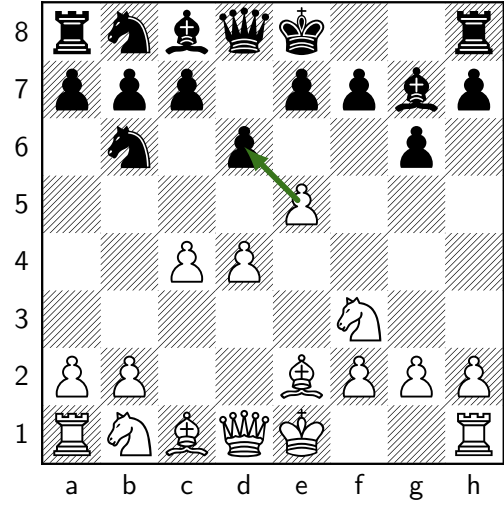


(d)

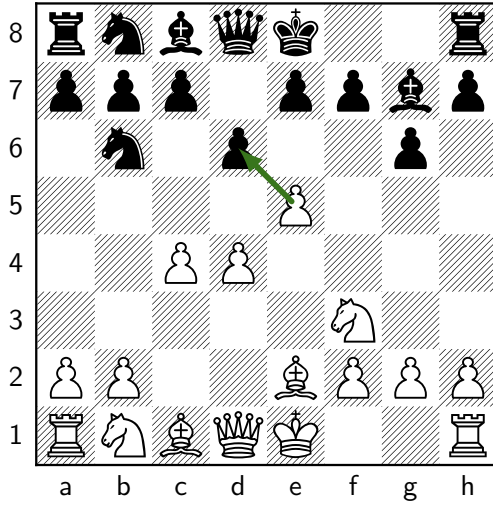
Figure 1: **Move 7, White: Rg1.** (a) Our position; Melanie plays Rg1 (h1g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#3785 (ply 9) [structural distance $d=48$, QAP obj=73]: the master played Bd6; under μ our Rh1 plays the role of that game’s Bf1 (a **cross-type** correspondence); cross-type roles: our pa7 as their Ra8, our Nf3 as their Rh8; transferred move h1g1 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#3786 (ply 9) [structural distance $d=48$, QAP obj=73]: the master played Bd6; under μ our Rh1 plays the role of that game’s Bf1 (a **cross-type** correspondence); cross-type roles: our pa7 as their Ra8, our Nf3 as their Rh8; transferred move h1g1 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#3772 (ply 9) [structural distance $d=48$, QAP obj=73]: the master played Bd6; under μ our Rh1 plays the role of that game’s Bf1 (a **cross-type** correspondence); cross-type roles: our pa7 as their Ra8, our Nf3 as their Rh8; transferred move h1g1 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Rg1 (summed QAP objective 441, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



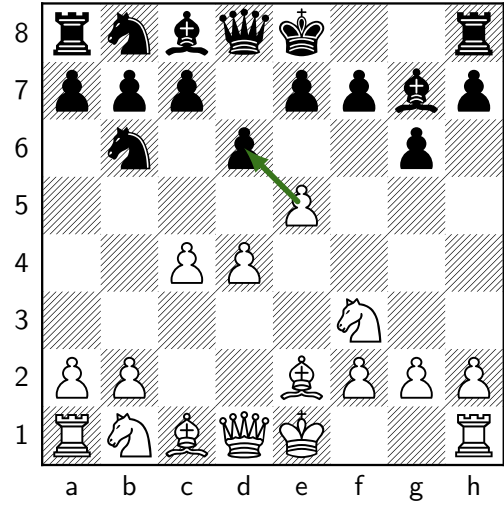
(a)



(b)

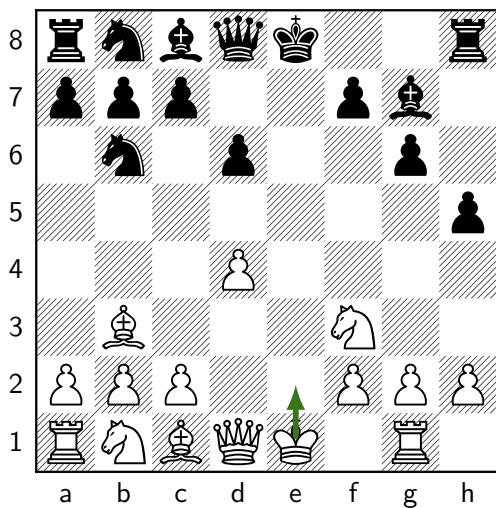


(c)

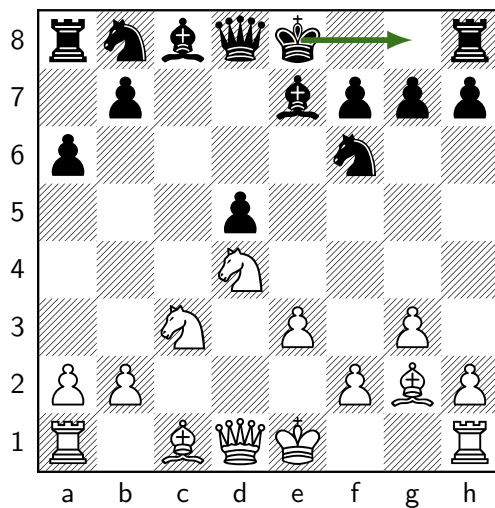


(d)

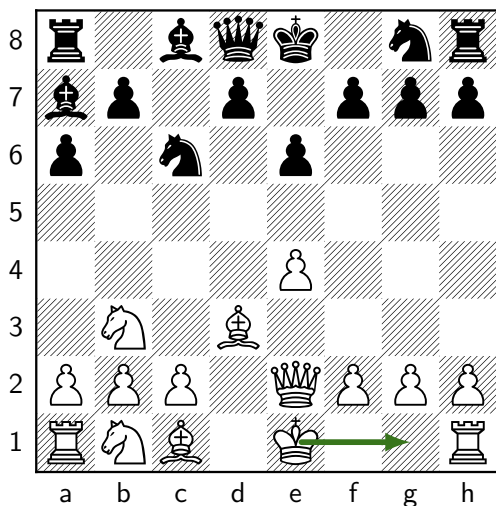
Figure 2: **Move 8, White: exd6.** (a) Our position; Melanie plays exd6 (e5d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#637 (ply 12) [structural distance $d=39$, QAP obj=78]: the master played exd6; under μ our Pe5 plays the role of that game’s Pe5; cross-type roles: our Pb2 as their Ra1, our Ra1 as their Nb1; transferred move e5d6 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#2419 (ply 12) [structural distance $d=39$, QAP obj=78]: the master played exd6; under μ our Pe5 plays the role of that game’s Pe5; cross-type roles: our Pb2 as their Ra1, our Ra1 as their Nb1; transferred move e5d6 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#1474 (ply 12) [structural distance $d=39$, QAP obj=78]: the master played exd6; under μ our Pe5 plays the role of that game’s Pe5; cross-type roles: our Pb2 as their Ra1, our Ra1 as their Nb1; transferred move e5d6 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to exd6 (summed QAP objective 519, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



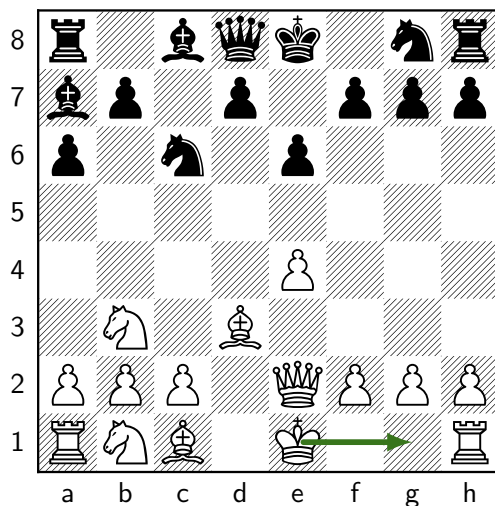
(a)



(b)

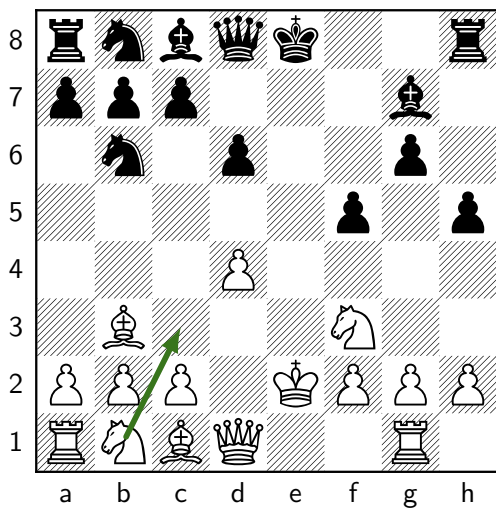


(c)

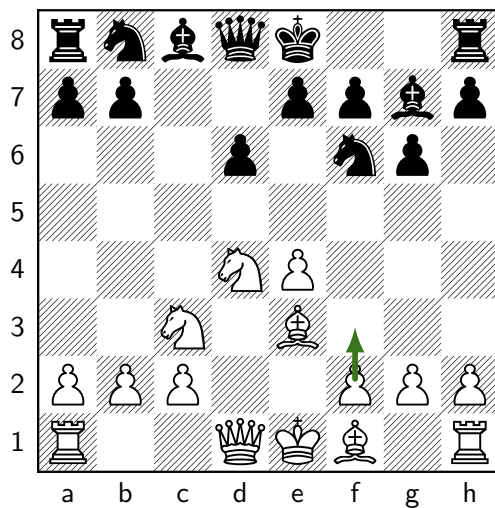


(d)

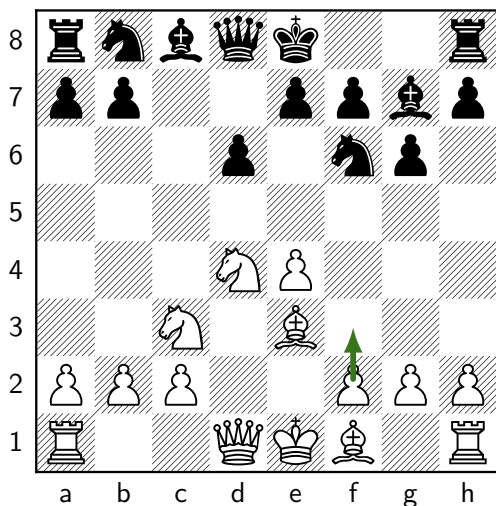
Figure 3: **Move 9, White: Ke2.** (a) Our position; Melanie plays Ke2 (e1e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2385` (ply 17) [structural distance $d=46$, QAP obj=38]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *pb7* as their *Nb8*, our *bg7* as their *nc3*; transferred move e1e2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1108` (ply 14) [structural distance $d=57$, QAP obj=33]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Qd1* as their *Ra1*, our *Ra1* as their *Nb1*; transferred move e1e2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1109` (ply 14) [structural distance $d=57$, QAP obj=33]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Qd1* as their *Ra1*, our *Ra1* as their *Nb1*; transferred move e1e2 (**endorses this move**). Synthesis: 14 retrieved analogs transferred to Ke2 (summed QAP objective 396, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



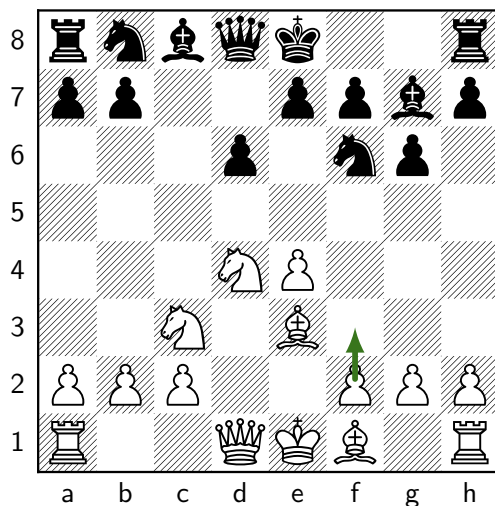
(a)



(b)

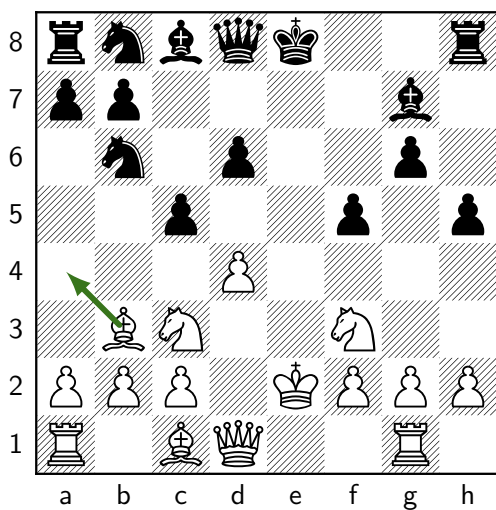


(c)

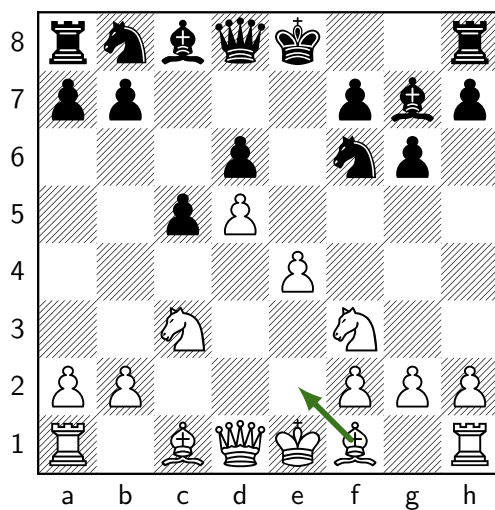


(d)

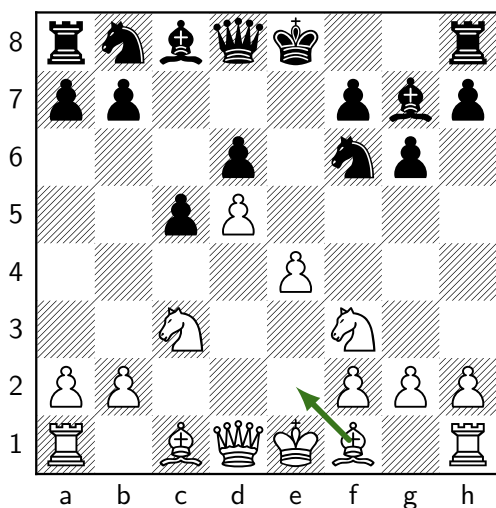
Figure 4: **Move 10, White: Nc3**. (a) Our position; Melanie plays Nc3 (b1c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#212 (ply 12) [structural distance $d=45$, QAP obj=18]: the master played f3; under μ our Nb1 plays the role of that game’s Pf2 (a **cross-type** correspondence); cross-type roles: our Bc1 as their Ra1, our pd6 as their Bf1; transferred move b1c3 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#213 (ply 12) [structural distance $d=45$, QAP obj=18]: the master played f3; under μ our Nb1 plays the role of that game’s Pf2 (a **cross-type** correspondence); cross-type roles: our Bc1 as their Ra1, our pd6 as their Bf1; transferred move b1c3 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#298 (ply 12) [structural distance $d=45$, QAP obj=18]: the master played f3; under μ our Nb1 plays the role of that game’s Pf2 (a **cross-type** correspondence); cross-type roles: our Bc1 as their Ra1, our pd6 as their Bf1; transferred move b1c3 (**endorses this move**). Synthesis: 56 retrieved analogs transferred to Nc3 (summed QAP objective 1018, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



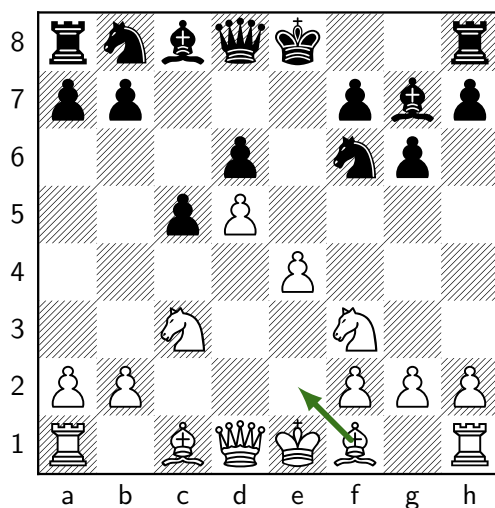
(a)



(b)

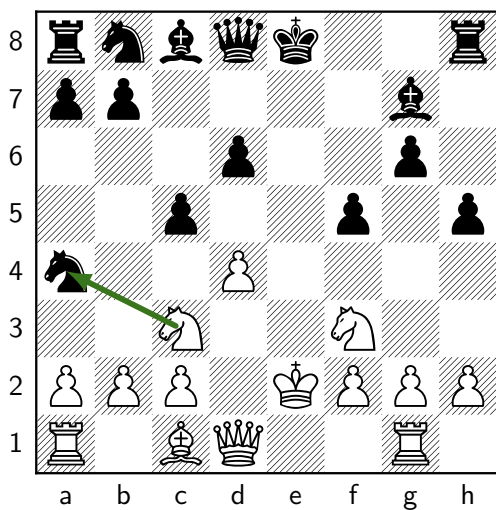


(c)

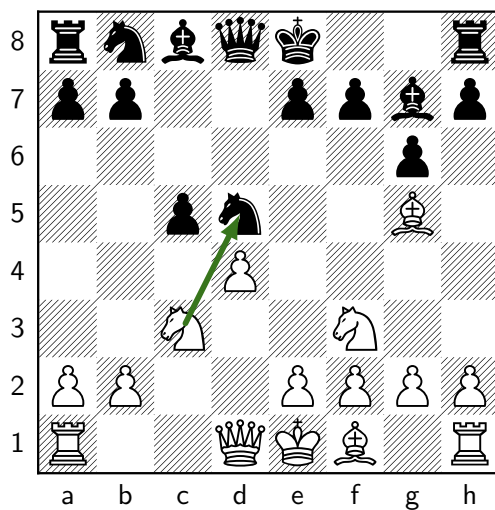


(d)

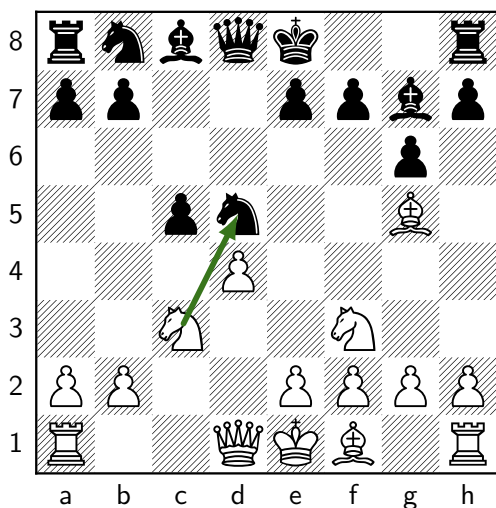
Figure 5: **Move 11, White: Ba4+.** (a) Our position; Melanie plays Ba4+ (b3a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0000#1801` (ply 14) [structural distance $d=56$, QAP obj=21]: the master played Be2; under μ our Bb3 plays the role of that game’s Bf1; cross-type roles: our Qd1 as their Ra1, our Nf3 as their Bc1; transferred move b3a4 (**endorses this move**). (c) Retrieved from `ccl1_top_p0000#1802` (ply 14) [structural distance $d=56$, QAP obj=21]: the master played Be2; under μ our Bb3 plays the role of that game’s Bf1; cross-type roles: our Qd1 as their Ra1, our Nf3 as their Bc1; transferred move b3a4 (**endorses this move**). (d) Retrieved from `ccl1_top_p0000#2113` (ply 14) [structural distance $d=56$, QAP obj=21]: the master played Be2; under μ our Bb3 plays the role of that game’s Bf1; cross-type roles: our Qd1 as their Ra1, our Nf3 as their Bc1; transferred move b3a4 (**endorses this move**). Synthesis: 18 retrieved analogs transferred to Ba4+ (summed QAP objective 374, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



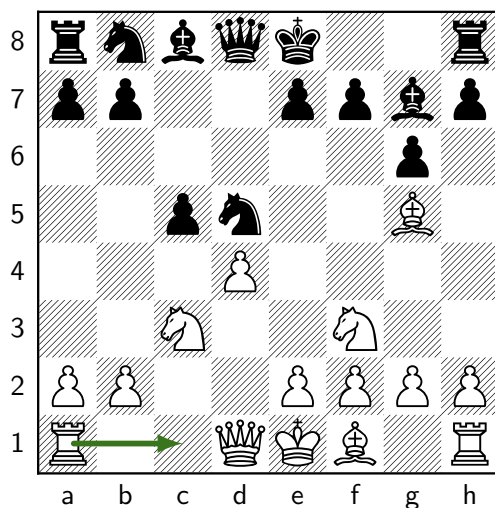
(a)



(b)

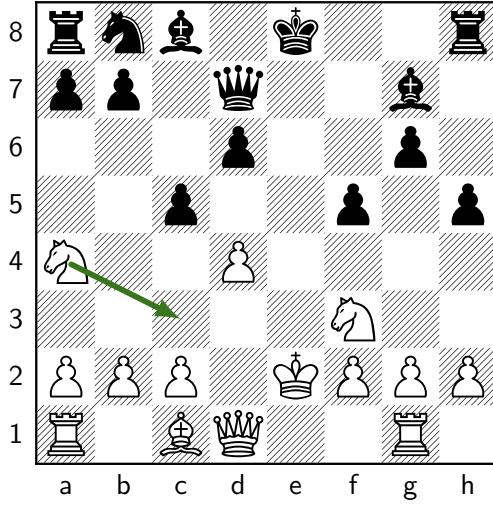


(c)

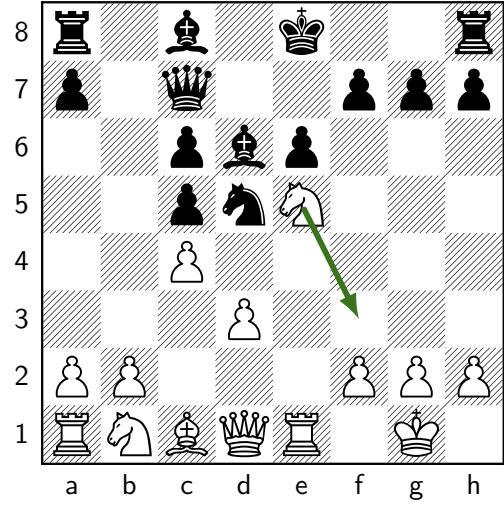


(d)

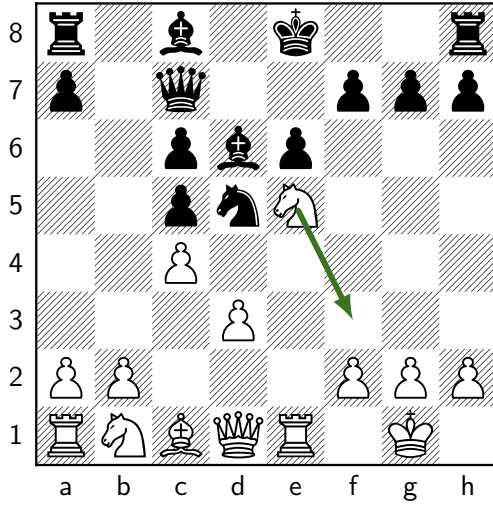
Figure 6: **Move 12, White: Nxa4.** (a) Our position; Melanie plays Nxa4 (c3a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1619` (ply 12) [structural distance $d=71$, QAP obj=109]: the master played Nxd5; under μ our Nc3 plays the role of that game’s Nc3; cross-type roles: our Pg2 as their Bf1, our nb8 as their Rh1; transferred move c3e4. (c) Retrieved from `ccr1_top_p0001#1620` (ply 12) [structural distance $d=71$, QAP obj=109]: the master played Nxd5; under μ our Nc3 plays the role of that game’s Nc3; cross-type roles: our Pg2 as their Bf1, our nb8 as their Rh1; transferred move c3e4. (d) Retrieved from `ccr1_top_p0001#1899` (ply 12) [structural distance $d=71$, QAP obj=109]: the master played Rc1; under μ our Rg1 plays the role of that game’s Ra1; cross-type roles: our Pg2 as their Bf1, our nb8 as their Rh1; transferred move g1h1. Synthesis: no retrieved analog transferred to Nxa4 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



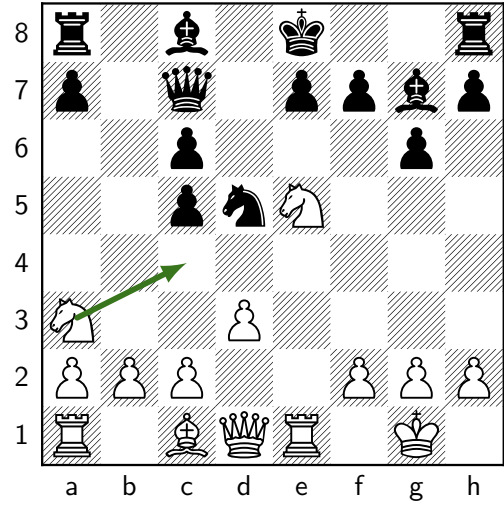
(a)



(b)

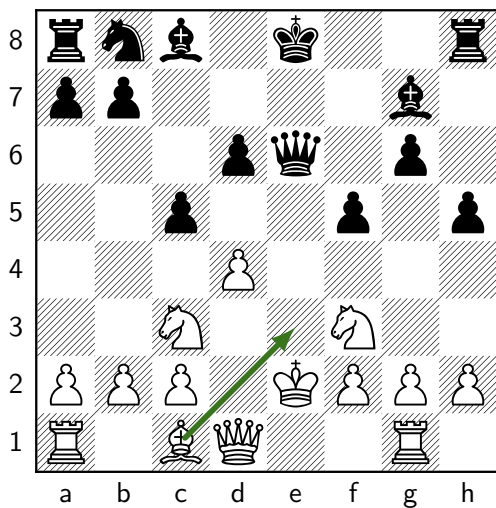


(c)

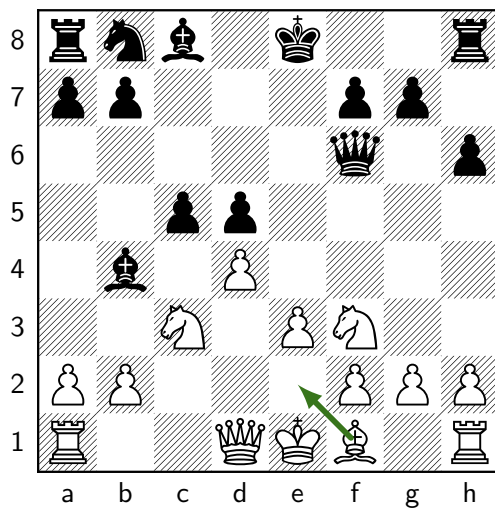


(d)

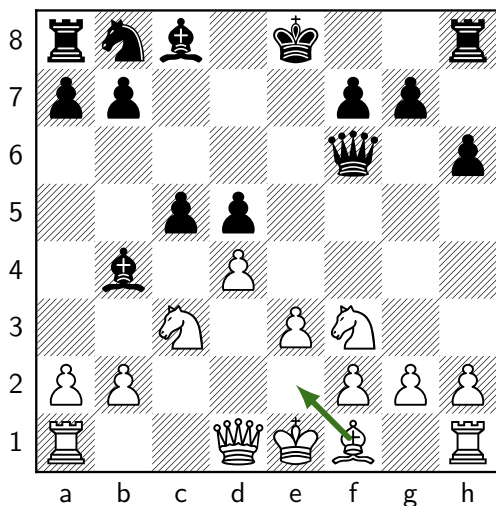
Figure 7: **Move 13, White: Nc3.** (a) Our position; Melanie plays Nc3 (a4c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1197` (ply 22) [structural distance $d=68$, QAP obj=25]: the master played Nf3; under μ our Na4 plays the role of that game’s Ne5; cross-type roles: our pg6 as their Ra1, our Ra1 as their Nb1; transferred move a4c3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1198` (ply 22) [structural distance $d=68$, QAP obj=25]: the master played Nf3; under μ our Na4 plays the role of that game’s Ne5; cross-type roles: our pg6 as their Ra1, our Ra1 as their Nb1; transferred move a4c3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#593` (ply 20) [structural distance $d=76$, QAP obj=73]: the master played Nac4; under μ our Nf3 plays the role of that game’s Na3; cross-type roles: our Qd1 as their Bc1, our Pd4 as their Qd1; transferred move f3g5. Synthesis: 2 retrieved analogs transferred to Nc3 (summed QAP objective 50, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



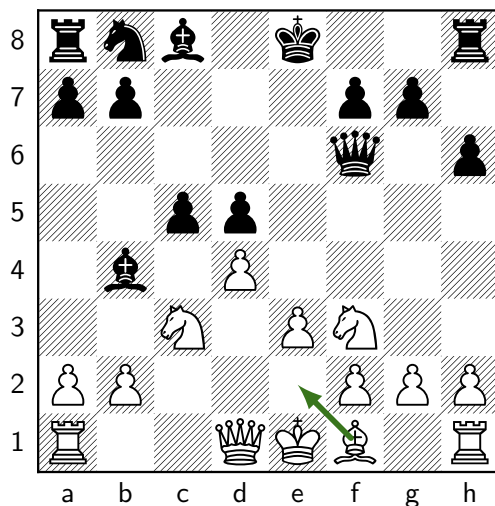
(a)



(b)

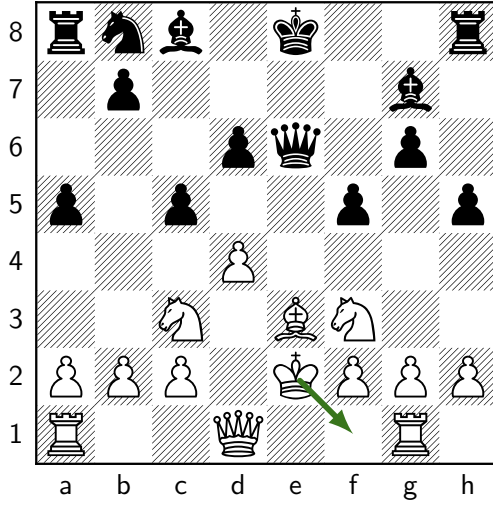


(c)

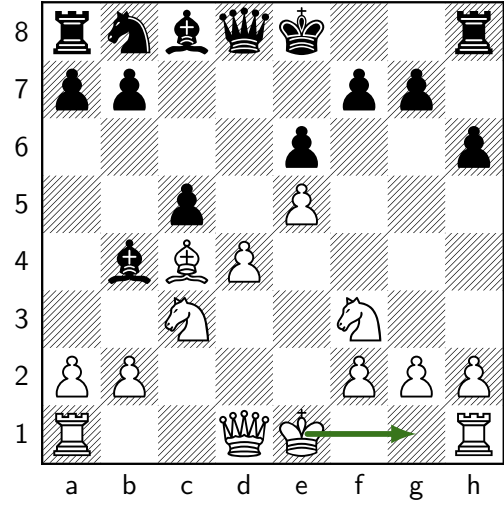


(d)

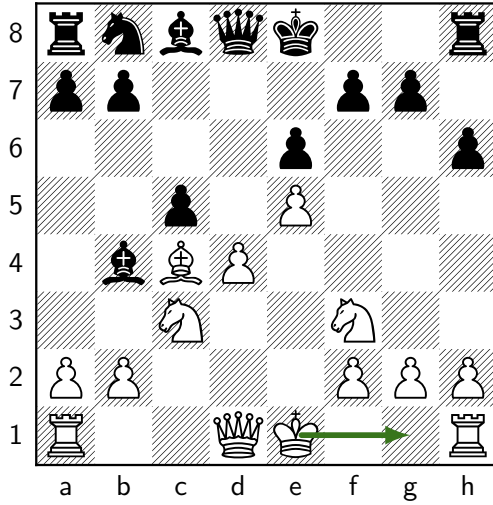
Figure 8: **Move 14, White: Be3 (c1e3).** (a) Our position; Melanie plays Be3 (c1e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3044` (ply 16) [structural distance $d=63$, QAP obj=20]: the master played Be2; under μ our Bc1 plays the role of that game’s Bf1; cross-type roles: our pd6 as their Ra1, our Nf3 as their Qd1; transferred move c1e3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3045` (ply 16) [structural distance $d=63$, QAP obj=20]: the master played Be2; under μ our Bc1 plays the role of that game’s Bf1; cross-type roles: our pd6 as their Ra1, our Nf3 as their Qd1; transferred move c1e3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3062` (ply 16) [structural distance $d=63$, QAP obj=20]: the master played Be2; under μ our Bc1 plays the role of that game’s Bf1; cross-type roles: our pd6 as their Ra1, our Nf3 as their Qd1; transferred move c1e3 (**endorses this move**). Synthesis: 13 retrieved analogs transferred to Be3 (summed QAP objective 236, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



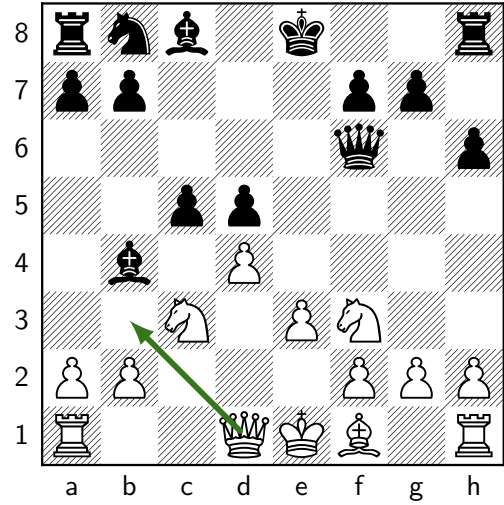
(a)



(b)

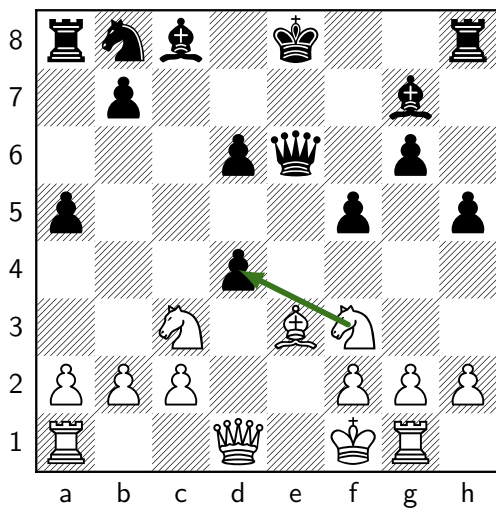


(c)

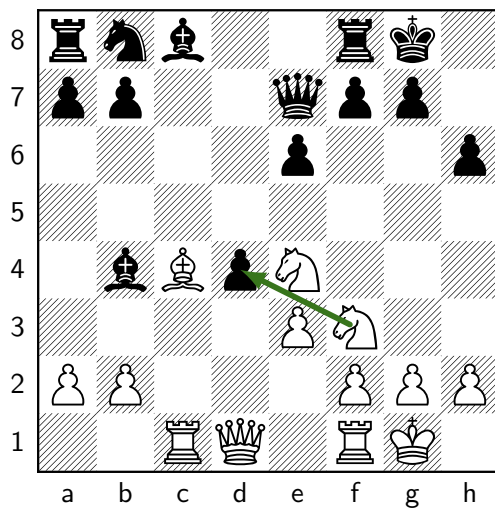


(d)

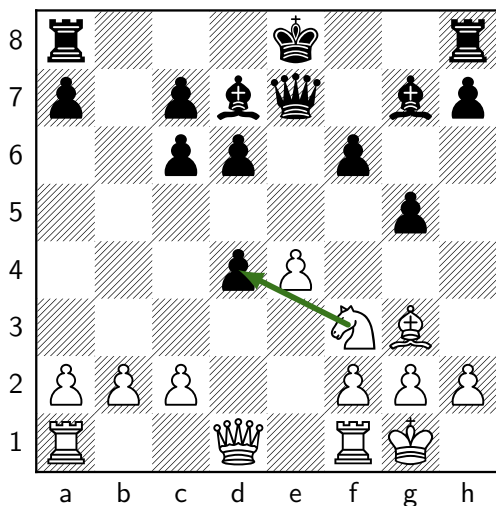
Figure 9: **Move 15, White: Kf1**. (a) Our position; Melanie plays Kf1 (e2f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#719` (ply 18) [structural distance $d=75$, QAP obj=54]: the master played 0-0; under μ our *Ke2* plays the role of that game’s *Ke1*; cross-type roles: our *Nc3* as their *Ra1*, our *nb8* as their *Ph2*; transferred move e2f1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#720` (ply 18) [structural distance $d=75$, QAP obj=54]: the master played 0-0; under μ our *Ke2* plays the role of that game’s *Ke1*; cross-type roles: our *Nc3* as their *Ra1*, our *nb8* as their *Ph2*; transferred move e2f1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3218` (ply 16) [structural distance $d=74$, QAP obj=46]: the master played Qb3; under μ our *Qd1* plays the role of that game’s *Qd1*; cross-type roles: our *pb7* as their *Bf1*, our *Nf3* as their *Rh1*; transferred move d1c1. Synthesis: 2 retrieved analogs transferred to Kf1 (summed QAP objective 107, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



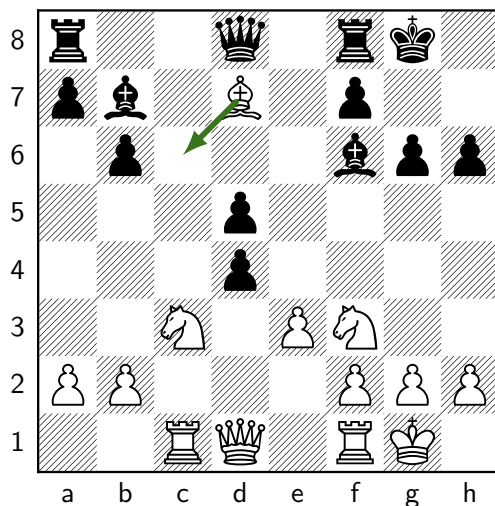
(a)



(b)

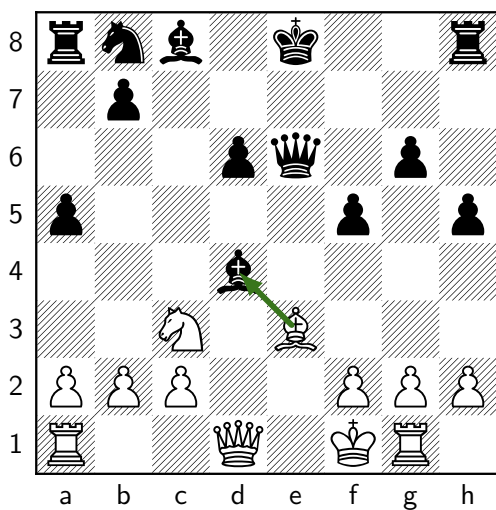


(c)

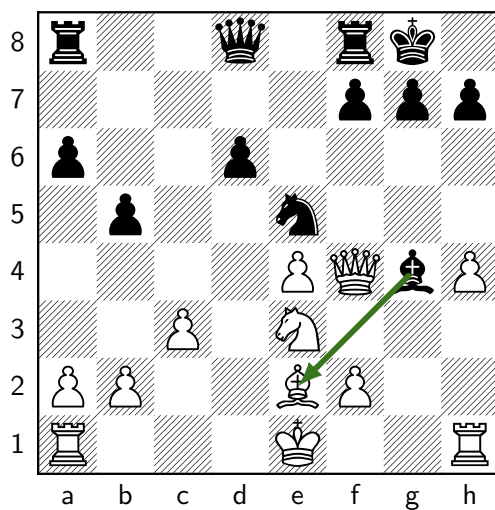


(d)

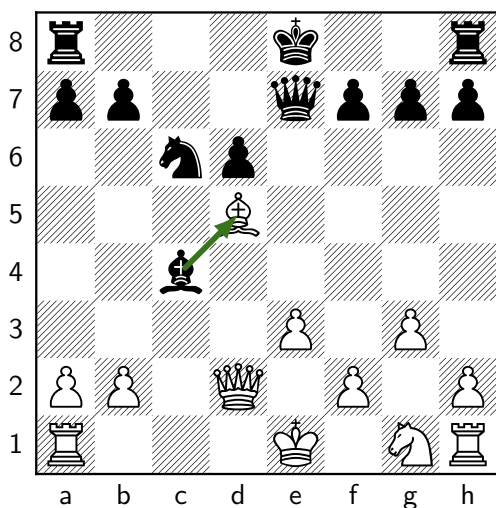
Figure 10: **Move 16, White: Nxd4.** (a) Our position; Melanie plays Nxd4 (f3d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2328` (ply 22) [structural distance $d=79$, QAP obj=39]: the master played Nxd4; under μ our *Nf3* plays the role of that game’s *Nf3*; cross-type roles: our *Pb2* as their *Rc1*, our *nb8* as their *Ph2*; transferred move f3d4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2979` (ply 24) [structural distance $d=87$, QAP obj=15]: the master played Nxd4; under μ our *Nf3* plays the role of that game’s *Nf3*; cross-type roles: our *Nc3* as their *Ra1*, our *Ra1* as their *Pa2*; transferred move f3d4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3899` (ply 28) [structural distance $d=85$, QAP obj=103]: the master played Bc6; under μ our *Be3* plays the role of that game’s *Bd7*; cross-type roles: our *Pc2* as their *Rc1*, our *Be3* as their *Pe3*; transferred move e3h6. Synthesis: 2 retrieved analogs transferred to Nxd4 (summed QAP objective 54, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



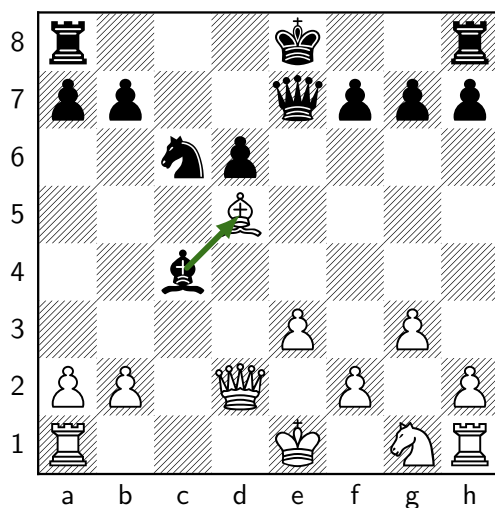
(a)



(b)

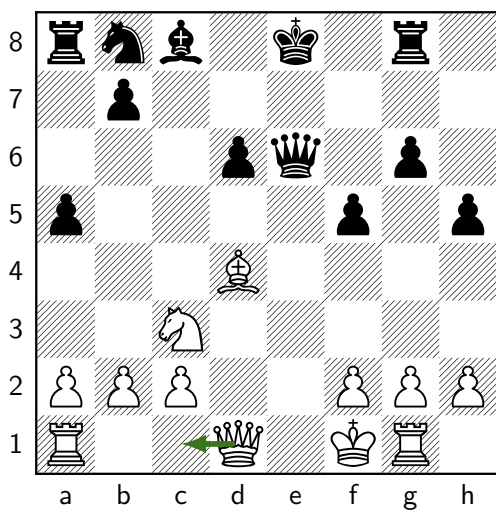


(c)

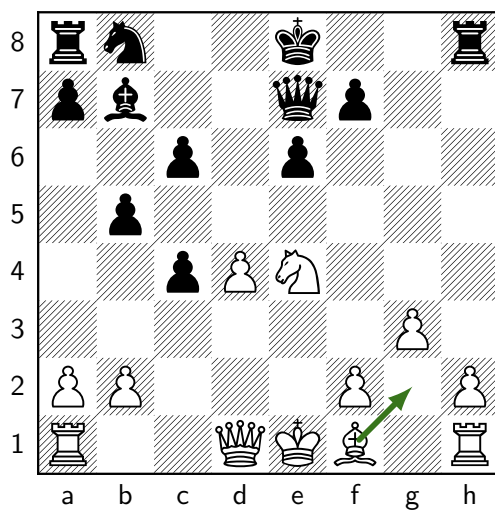


(d)

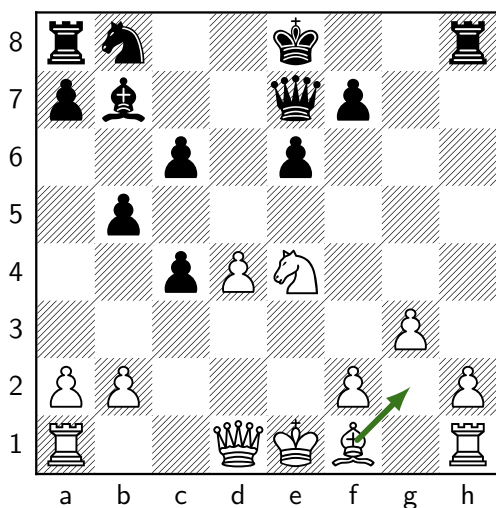
Figure 11: **Move 17, White: Bxd4.** (a) Our position; Melanie plays Bxd4 (e3d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1149` (ply 37) [structural distance $d=78$, QAP obj=89]: the master played Bxe2; under μ our *Be3* plays the role of that game’s *Bg5*; cross-type roles: our *nb8* as their *Ra8*, our *bd4* as their *qf4*; transferred move e3d4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2638` (ply 25) [structural distance $d=66$, QAP obj=80]: the master played Bxd5; under μ our *Be3* plays the role of that game’s *Bc5*; cross-type roles: our *qe6* as their *Ph7*, our *bc8* as their *pb2*; transferred move e3d4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2639` (ply 25) [structural distance $d=66$, QAP obj=80]: the master played Bxd5; under μ our *Be3* plays the role of that game’s *Bc5*; cross-type roles: our *qe6* as their *Ph7*, our *bc8* as their *pb2*; transferred move e3d4 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Bxd4 (summed QAP objective 426, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



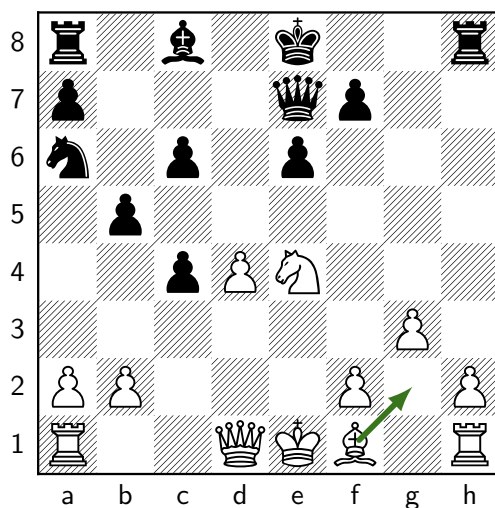
(a)



(b)

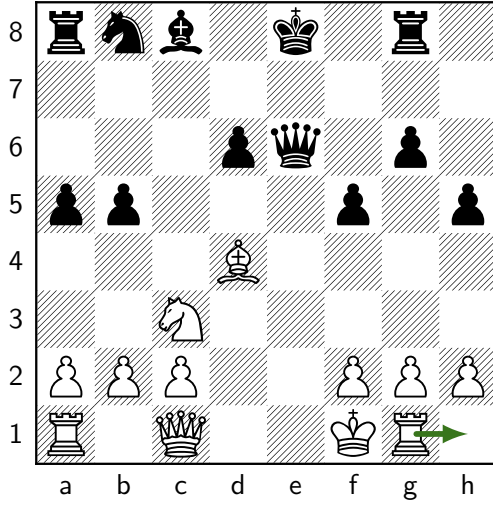


(c)

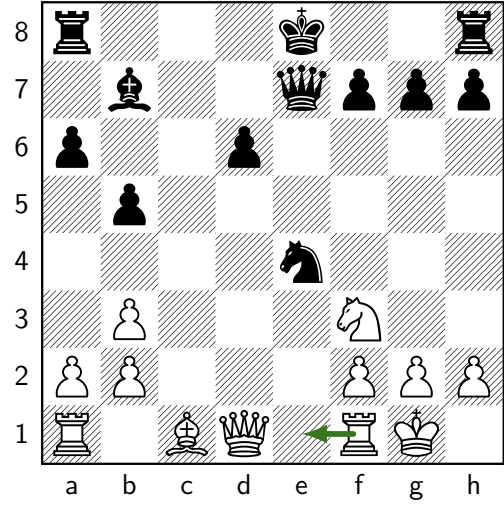


(d)

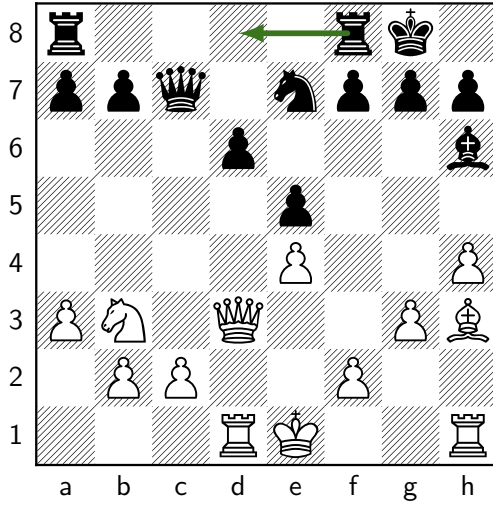
Figure 12: **Move 18, White: Qc1**. (a) Our position; Melanie plays Qc1 (d1c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3603` (ply 28) [structural distance $d=76$, QAP obj=30]: the master played Bg2; under μ our Qd1 plays the role of that game's Bf1 (a **cross-type** correspondence); cross-type roles: our Bd4 as their Ra1, our Rg1 as their Qd1; transferred move d1c1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1545` (ply 28) [structural distance $d=76$, QAP obj=30]: the master played Bg2; under μ our Qd1 plays the role of that game's Bf1 (a **cross-type** correspondence); cross-type roles: our Bd4 as their Ra1, our Rg1 as their Qd1; transferred move d1c1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#568` (ply 28) [structural distance $d=81$, QAP obj=30]: the master played Bg2; under μ our Qd1 plays the role of that game's Bf1 (a **cross-type** correspondence); cross-type roles: our pf5 as their Ra1, our Rg1 as their Qd1; transferred move d1c1 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Qc1 (summed QAP objective 151, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



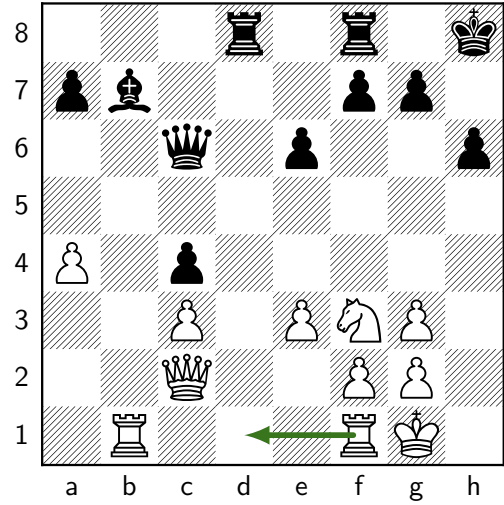
(a)



(b)

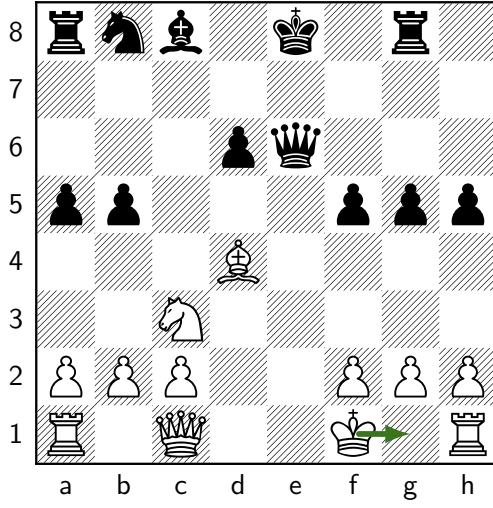


(c)

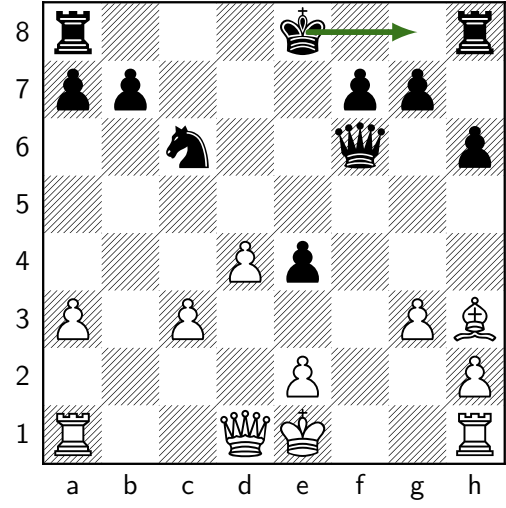


(d)

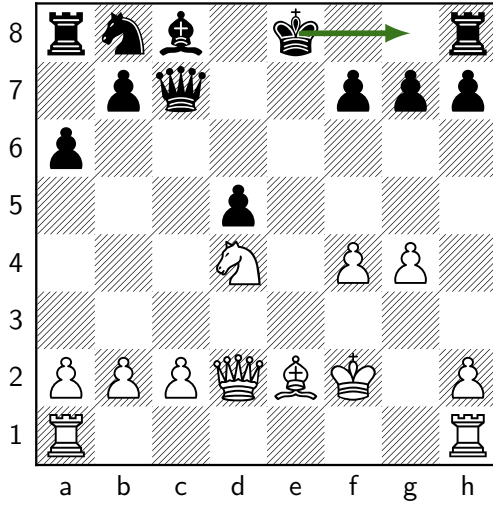
Figure 13: **Move 19, White: Rh1.** (a) Our position; Melanie plays Rh1 (**g1h1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#47` (ply 26) [structural distance $d=75$, QAP obj=34]: the master played **Re1**; under μ our *Rg1* plays the role of that game’s *Rf1*; cross-type roles: our *pf5* as their *Bc1*, our *nb8* as their *Pb2*; transferred move **g1h1** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3011` (ply 37) [structural distance $d=84$, QAP obj=26]: the master played **Rfd8**; under μ our *Rg1* plays the role of that game’s *Rf1*; cross-type roles: our *Pb2* as their *Ra8*, our *Qc1* as their *Pa7*; transferred move **g1h1** (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#359` (ply 44) [structural distance $d=85$, QAP obj=24]: the master played **Rfd1**; under μ our *Rg1* plays the role of that game’s *Rf1*; cross-type roles: our *qe6* as their *Rb1*, our *bc8* as their *Qc2*; transferred move **g1h1** (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Rh1 (summed QAP objective 137, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



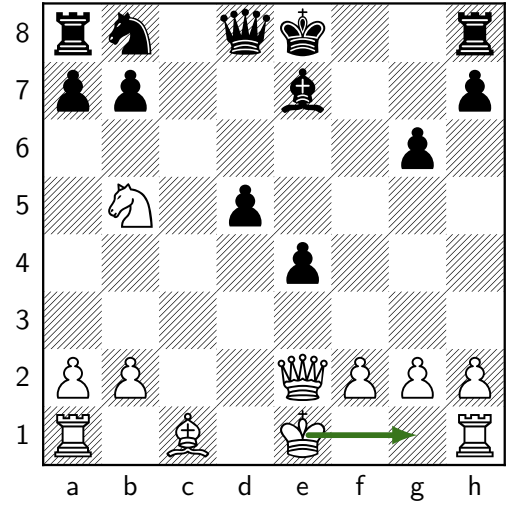
(a)



(b)

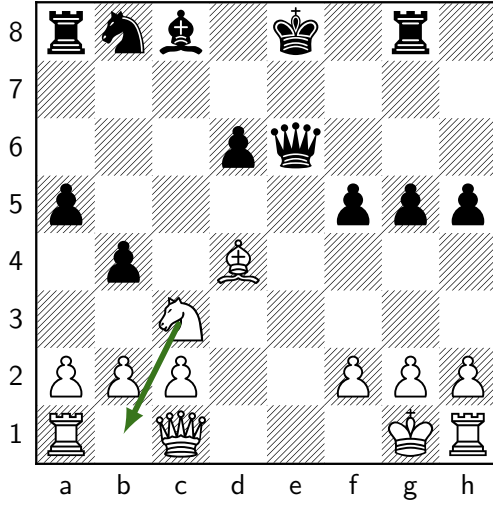


(c)

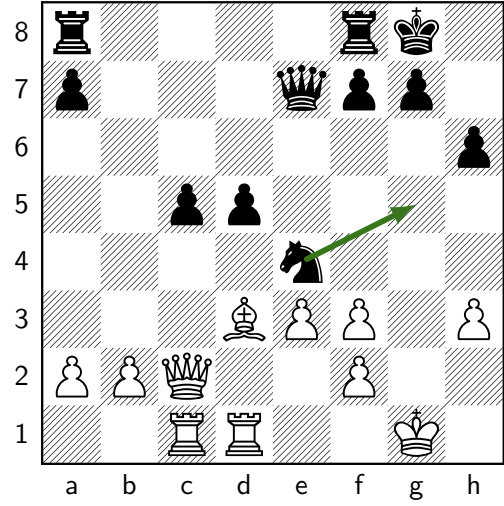


(d)

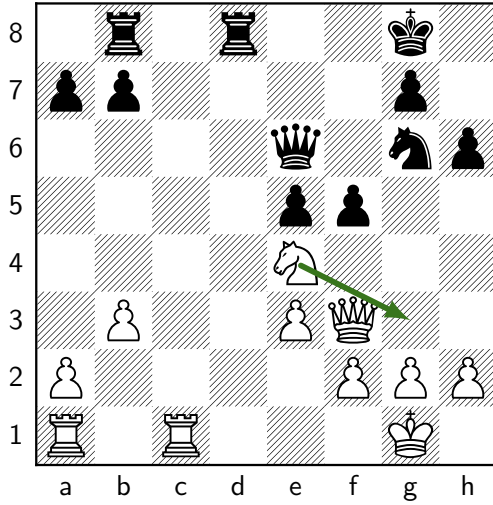
Figure 14: **Move 20, White: Kg1**. (a) Our position; Melanie plays Kg1 (f1g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2908` (ply 27) [structural distance $d=85$, QAP obj=39]: the master played 0-0; under μ our *Kf1* plays the role of that game’s *Ke1*; cross-type roles: our *Qc1* as their *Rh8*, our *Bd4* as their *Pb7*; transferred move f1g1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1434` (ply 27) [structural distance $d=87$, QAP obj=38]: the master played 0-0; under μ our *Kf1* plays the role of that game’s *Ke1*; cross-type roles: our *Pg2* as their *Ra8*, our *Pc2* as their *Nb8*; transferred move f1g1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3092` (ply 28) [structural distance $d=71$, QAP obj=31]: the master played 0-0; under μ our *Kf1* plays the role of that game’s *Ke1*; cross-type roles: our *Pf2* as their *Ra1*, our *Ra1* as their *pb7*; transferred move f1g1 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Kg1 (summed QAP objective 278, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



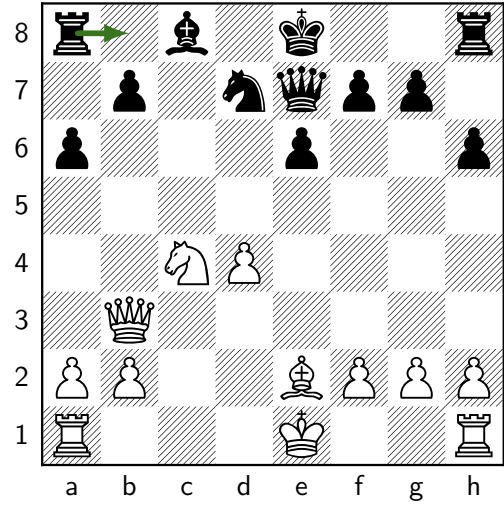
(a)



(b)

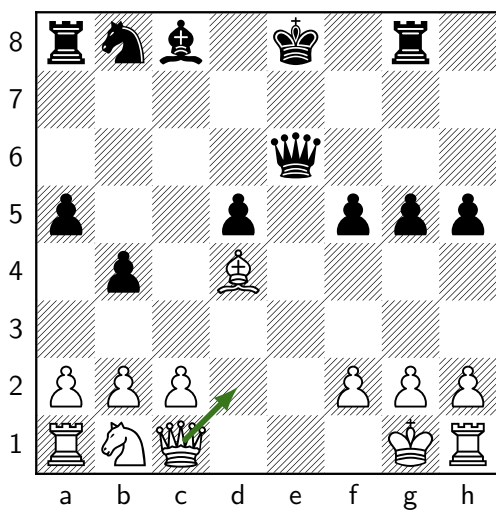


(c)

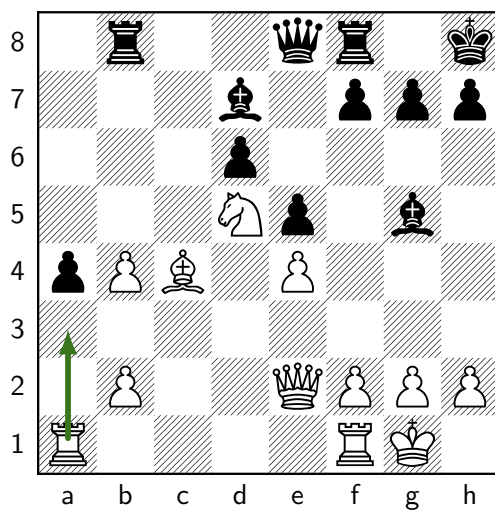


(d)

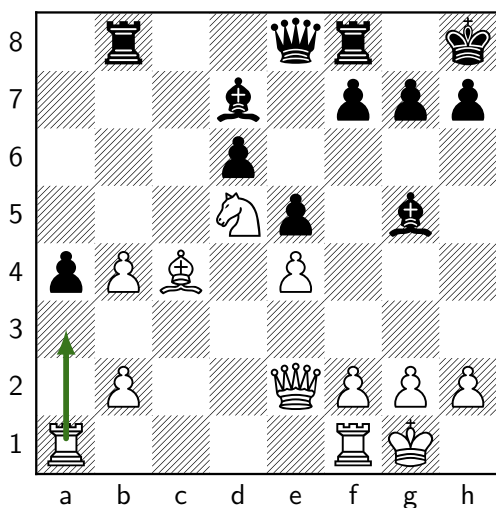
Figure 15: **Move 21, White: Nb1**. (a) Our position; Melanie plays Nb1 (c3b1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3196` (ply 39) [structural distance $d=83$, QAP obj=40]: the master played Ng5; under μ our Nc3 plays the role of that game’s Ne5; cross-type roles: our bc8 as their Ra8, our Bd4 as their Qe7; transferred move c3b1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#21` (ply 50) [structural distance $d=58$, QAP obj=22]: the master played Ng3; under μ our Nc3 plays the role of that game’s Ne4; cross-type roles: our ra8 as their Pb3, our Bd4 as their Pe3; transferred move c3b1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#878` (ply 29) [structural distance $d=81$, QAP obj=6]: the master played Rb8; under μ our Nc3 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Qc1 as their Rh8, our Pf2 as their Nd7; transferred move c3b1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Nb1 (summed QAP objective 69, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



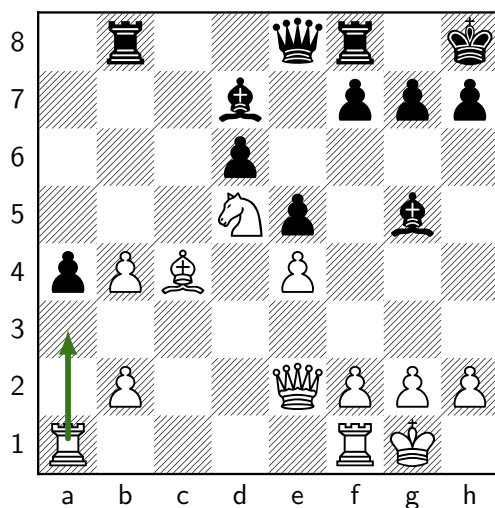
(a)



(b)

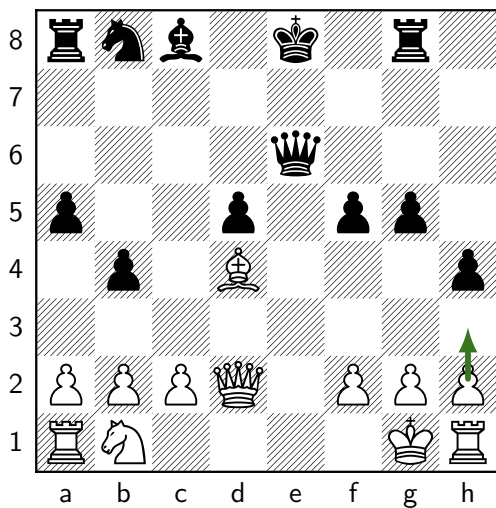


(c)

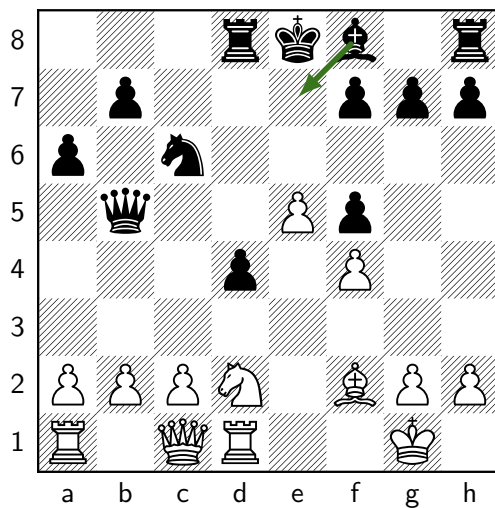


(d)

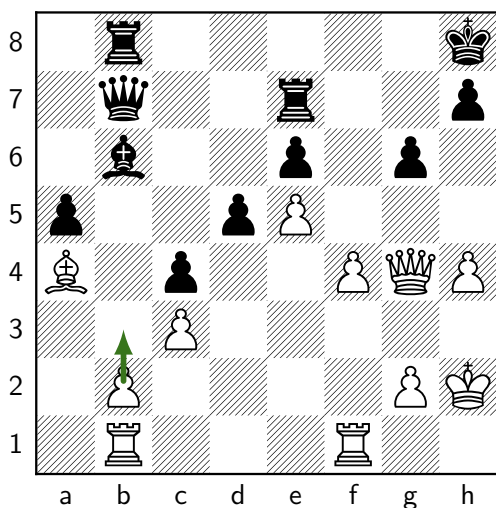
Figure 16: **Move 22, White: Qd2.** (a) Our position; Melanie plays Qd2 (c1d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#93` (ply 44) [structural distance $d=96$, QAP obj=23]: the master played Ra3; under μ our Qc1 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Ra1 as their Pb2, our Nb1 as their Qe2; transferred move c1d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#211` (ply 40) [structural distance $d=96$, QAP obj=23]: the master played Ra3; under μ our Qc1 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Ra1 as their Pb2, our Nb1 as their Qe2; transferred move c1d2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#91` (ply 38) [structural distance $d=96$, QAP obj=23]: the master played Ra3; under μ our Qc1 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Ra1 as their Pb2, our Nb1 as their Qe2; transferred move c1d2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qd2 (summed QAP objective 94, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



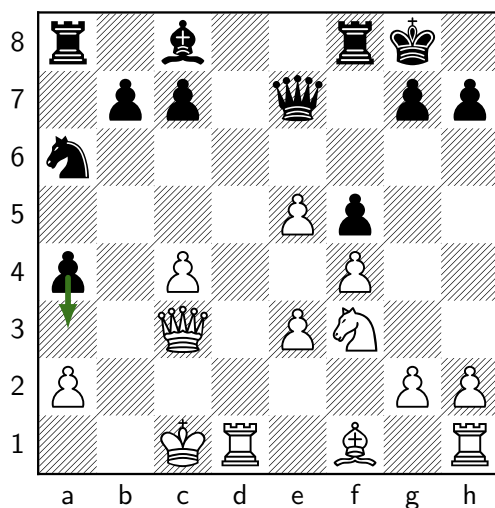
(a)



(b)

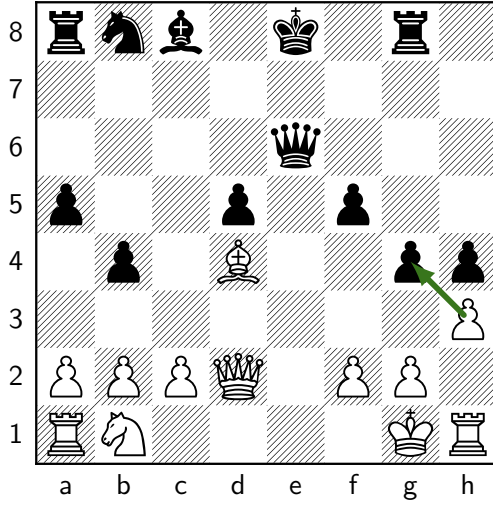


(c)

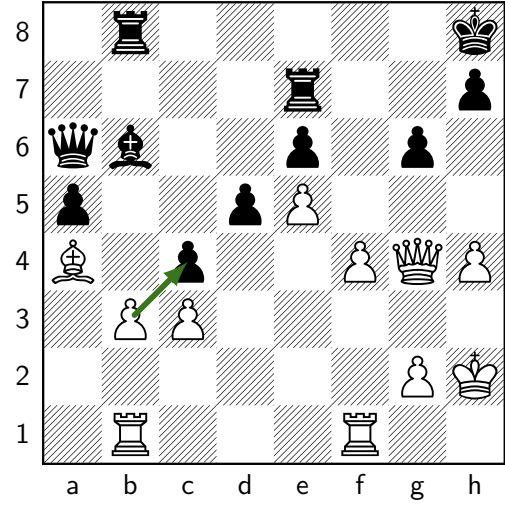


(d)

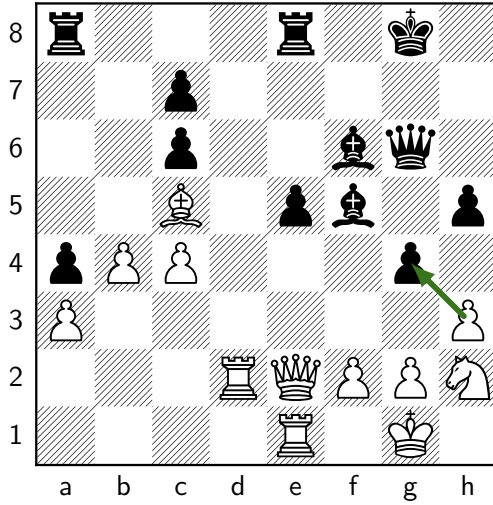
Figure 17: **Move 23, White: h3.** (a) Our position; Melanie plays h3 (h2h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2501` (ply 31) [structural distance $d=81$, QAP obj=54]: the master played Be7; under μ our *Ph2* plays the role of that game’s *Bf1* (a **cross-type** correspondence); cross-type roles: our *Nb1* as their *Pb7*, our *nb8* as their *Ph7*; transferred move h2h3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1297` (ply 56) [structural distance $d=83$, QAP obj=40]: the master played b3; under μ our *Ph2* plays the role of that game’s *Pb2*; cross-type roles: our *pg5* as their *Rb1*, our *qe6* as their *Rf1*; transferred move h2h3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#794` (ply 31) [structural distance $d=77$, QAP obj=27]: the master played a3; under μ our *Ph2* plays the role of that game’s *Pa5*; cross-type roles: our *Nb1* as their *Ra8*, our *Pa2* as their *Bc8*; transferred move h2h3 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to h3 (summed QAP objective 171, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



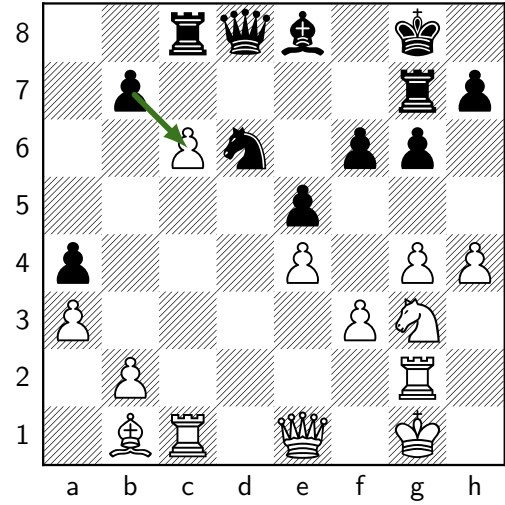
(a)



(b)

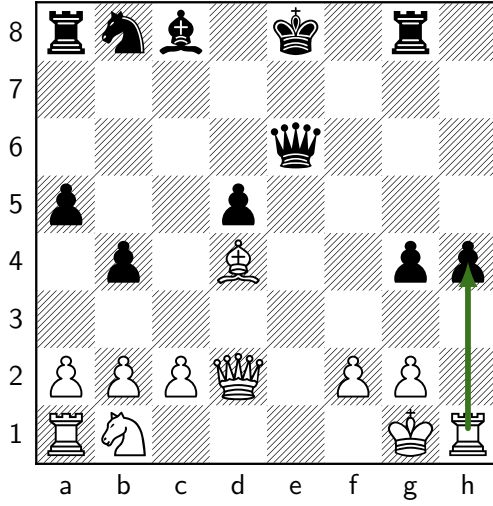


(c)

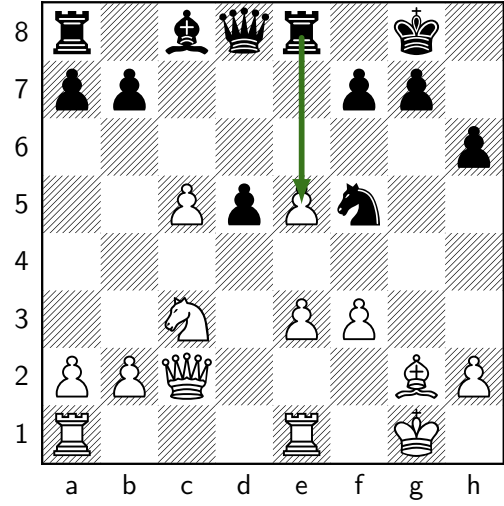


(d)

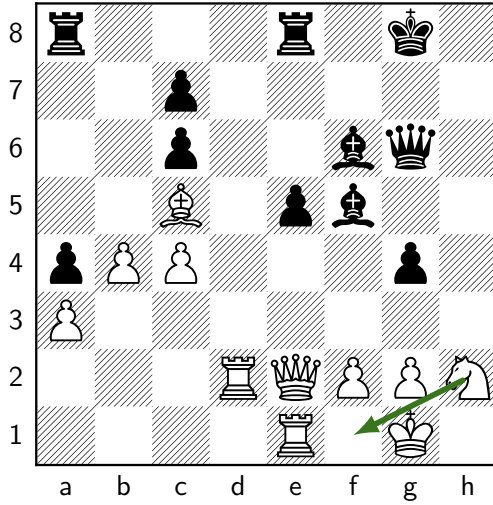
Figure 18: **Move 24, White: hxg4.** (a) Our position; Melanie plays hxg4 (h3g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#1297 (ply 58) [structural distance $d=76$, QAP obj=64]: the master played bxc4; under μ our Ph3 plays the role of that game’s Pb3; cross-type roles: our Pf2 as their Rb1, our Nb1 as their Pg2; transferred move h3g4 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#3869 (ply 50) [structural distance $d=85$, QAP obj=63]: the master played hxg4; under μ our Ph3 plays the role of that game’s Ph3; cross-type roles: our Pg2 as their Rd2, our Ph3 as their Qe2; transferred move h3g4 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#3776 (ply 63) [structural distance $d=80$, QAP obj=62]: the master played bxc6; under μ our Ph3 plays the role of that game’s Pb2; cross-type roles: our Ph3 as their Rc8, our Pg2 as their Qd8; transferred move h3g4 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to hxg4 (summed QAP objective 325, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



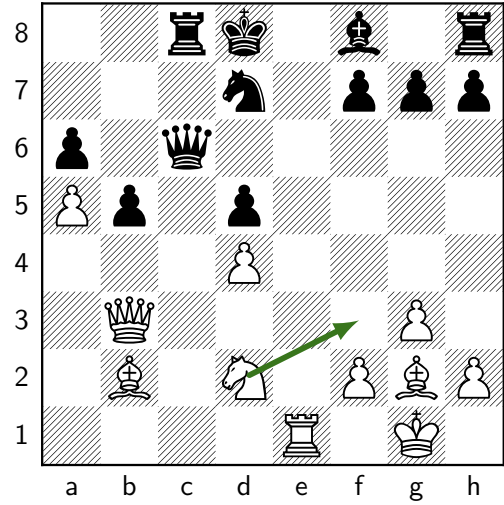
(a)



(b)

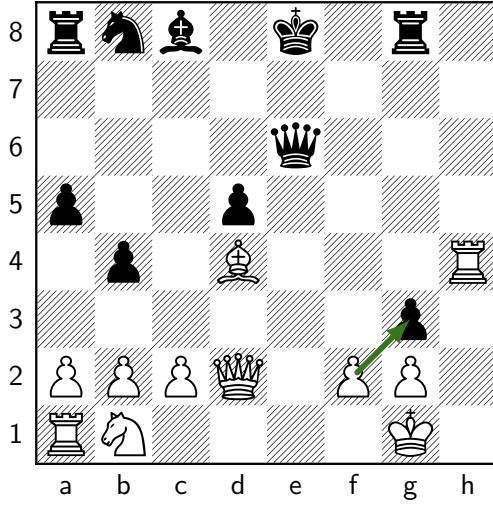


(c)

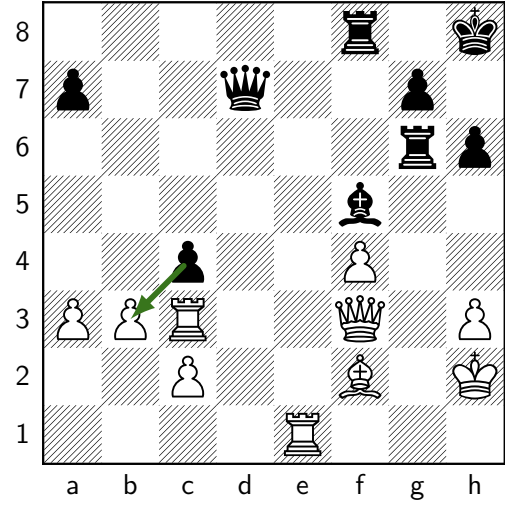


(d)

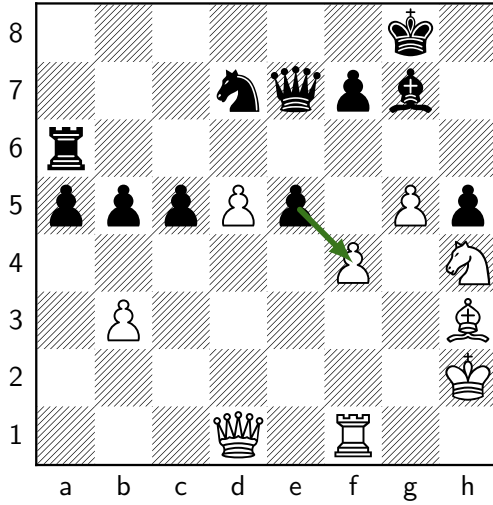
Figure 19: **Move 25, White: Rxe4**. (a) Our position; Melanie plays Rxe4 (h1h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2543` (ply 33) [structural distance $d=76$, QAP obj=21]: the master played Rxe5; under μ our *Rh1* plays the role of that game’s *Re1*; cross-type roles: our *pg4* as their *Bc8*, our *Bd4* as their *Qd8*; transferred move h1h4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3869` (ply 52) [structural distance $d=62$, QAP obj=53]: the master played Nf1; under μ our *Qd2* plays the role of that game’s *Nh2* (a **cross-type** correspondence); cross-type roles: our *Pc2* as their *Re1*, our *Nb1* as their *Rd2*; transferred move d2c1. (d) Retrieved from `ccr1_top_p0001#3745` (ply 38) [structural distance $d=86$, QAP obj=50]: the master played Nf3; under μ our *Nb1* plays the role of that game’s *Nd2*; cross-type roles: our *Pc2* as their *Bb2*, our *Qd2* as their *Bg2*; transferred move b1c3. Synthesis: 1 retrieved analog transferred to Rxe4 (summed QAP objective 21, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



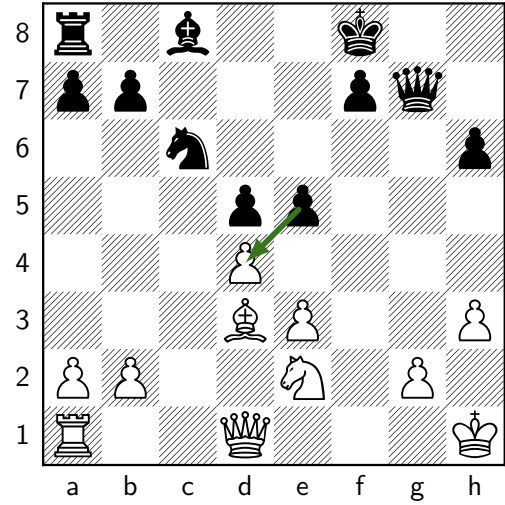
(a)



(b)

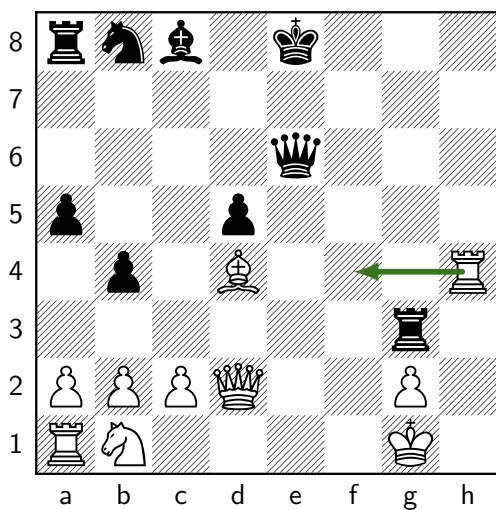


(c)

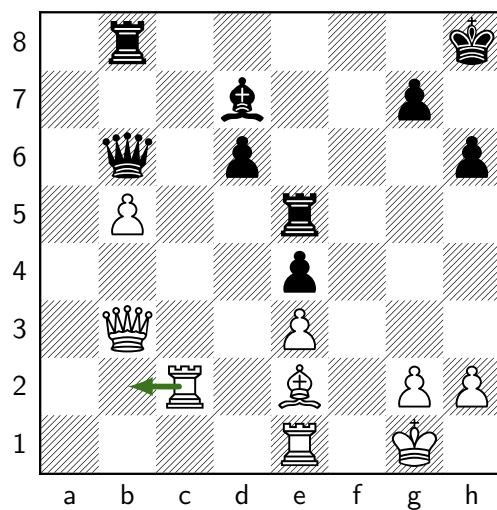


(d)

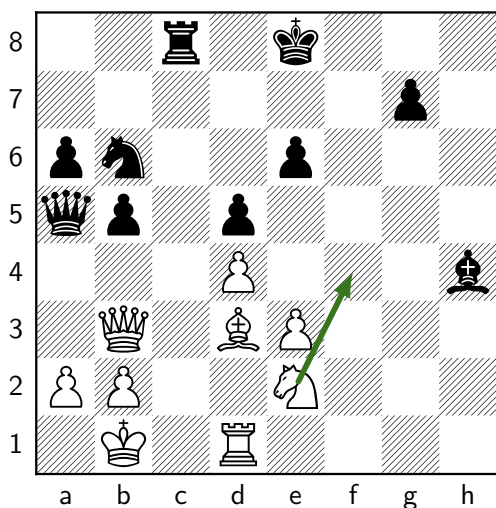
Figure 20: **Move 26, White: f2g3.** (a) Our position; Melanie plays f2g3 (f2g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#2259 (ply 61) [structural distance $d=88$, QAP obj=63]: the master played cxb3; under μ our Pf2 plays the role of that game’s Pc5; cross-type roles: our Nb1 as their Rf8, our qe6 as their Pa7; transferred move f2g3 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#3164 (ply 63) [structural distance $d=75$, QAP obj=57]: the master played exf4; under μ our Pf2 plays the role of that game’s Pe4; cross-type roles: our pg3 as their Nd7, our Nb1 as their Bg7; transferred move f2g3 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#1278 (ply 39) [structural distance $d=87$, QAP obj=45]: the master played exd4; under μ our Pf2 plays the role of that game’s Pe4; cross-type roles: our Nb1 as their Pb7, our nb8 as their Pf7; transferred move f2g3 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to f2g3 (summed QAP objective 188, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



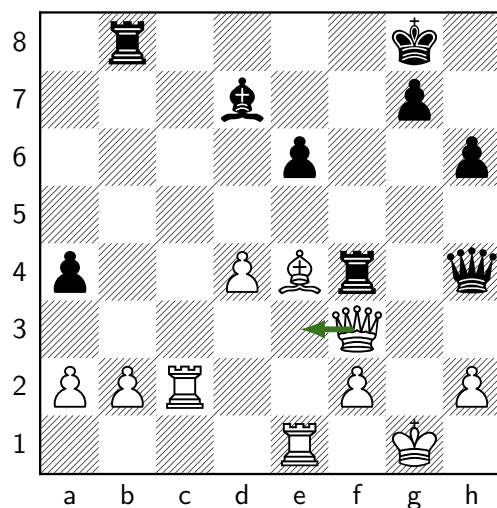
(a)



(b)

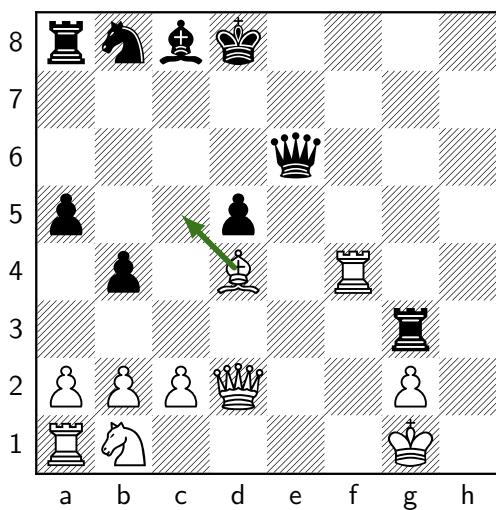


(c)

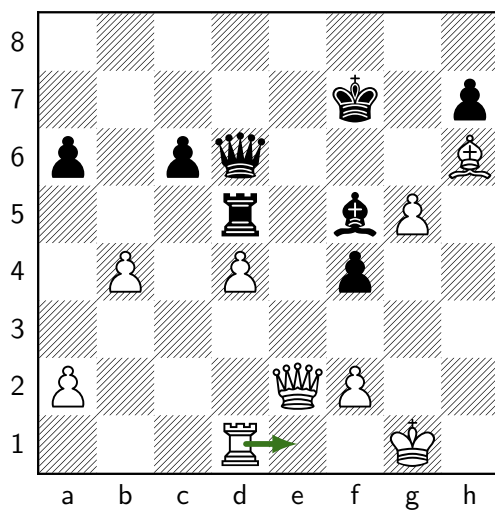


(d)

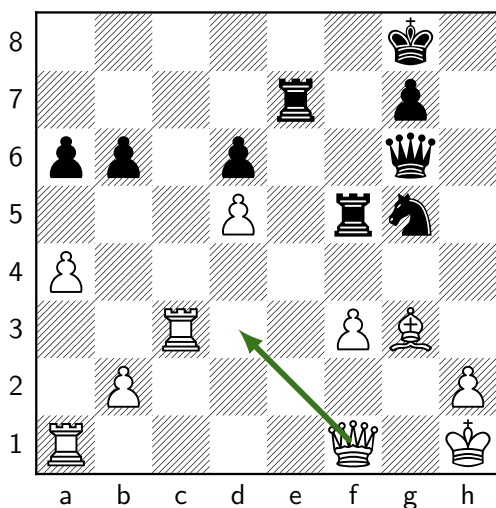
Figure 21: **Move 27, White: Rf4**. (a) Our position; Melanie plays Rf4 (**h4f4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#29` (ply 58) [structural distance $d=97$, QAP obj=61]: the master played Rb2; under μ our *Rh4* plays the role of that game’s *Rc2*; cross-type roles: our *Bd4* as their *Re1*, our *Pc2* as their *Be2*; transferred move **h4f4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3459` (ply 46) [structural distance $d=98$, QAP obj=43]: the master played Nf4; under μ our *Rh4* plays the role of that game’s *Ne2* (a **cross-type** correspondence); cross-type roles: our *Bd4* as their *Rd1*, our *rg3* as their *Qb3*; transferred move **h4f4** (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3650` (ply 56) [structural distance $d=98$, QAP obj=16]: the master played Qe3; under μ our *Rh4* plays the role of that game’s *Qf3* (a **cross-type** correspondence); cross-type roles: our *Bd4* as their *Re1*, our *ra8* as their *Pb2*; transferred move **h4f4** (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rf4 (summed QAP objective 182, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



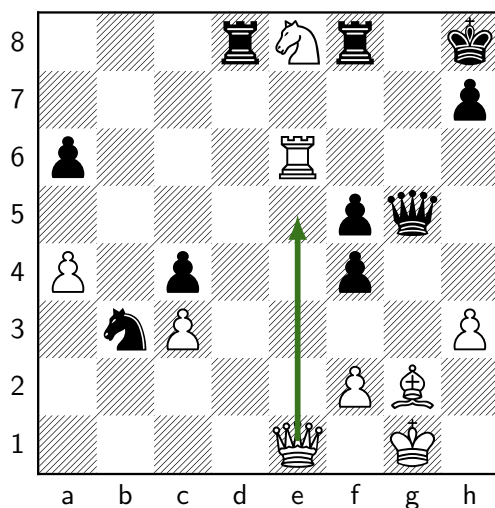
(a)



(b)

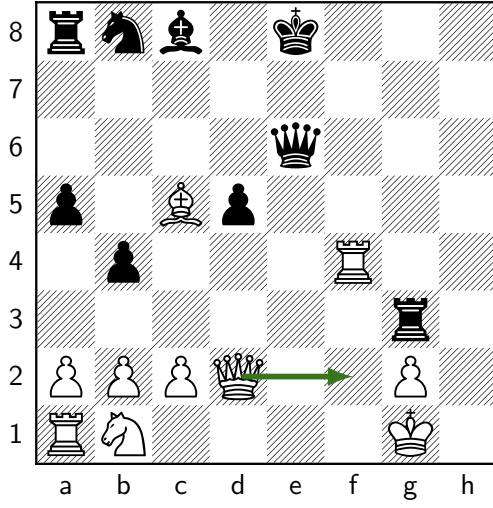


(c)

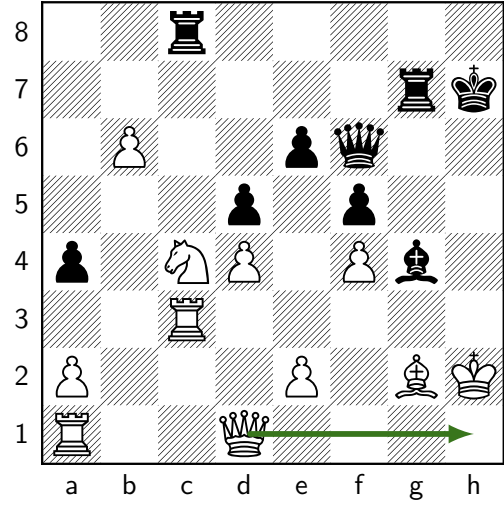


(d)

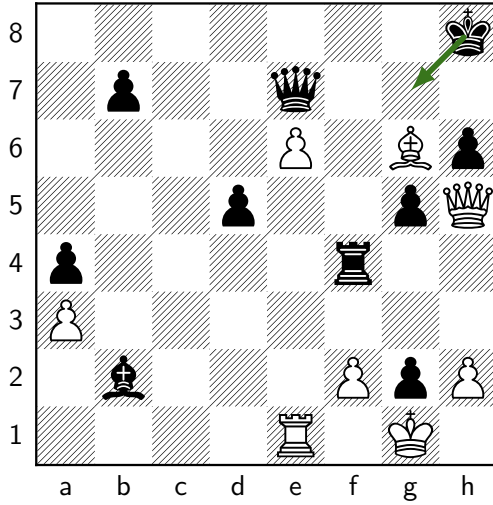
Figure 22: **Move 28, White: Bc5.** (a) Our position; Melanie plays Bc5 (d4c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2606` (ply 68) [structural distance $d=81$, QAP obj=58]: the master played Re1; under μ our Bd4 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Ra1 as their Pa2, our Rf4 as their Qe2; transferred move d4c5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#3809` (ply 64) [structural distance $d=97$, QAP obj=38]: the master played Qd3; under μ our Bd4 plays the role of that game’s Qf1 (a **cross-type** correspondence); cross-type roles: our Pb2 as their Ra1, our Nb1 as their Pb2; transferred move d4c5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1568` (ply 70) [structural distance $d=94$, QAP obj=31]: the master played Qe5+; under μ our Bd4 plays the role of that game’s Qe1 (a **cross-type** correspondence); cross-type roles: our nb8 as their Pf2, our Pg2 as their Bg2; transferred move d4c5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bc5 (summed QAP objective 137, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



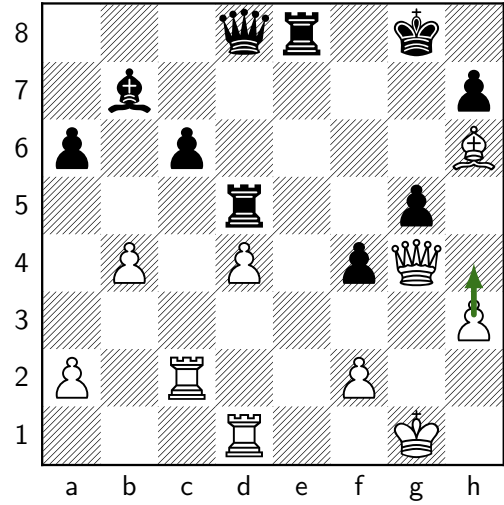
(a)



(b)

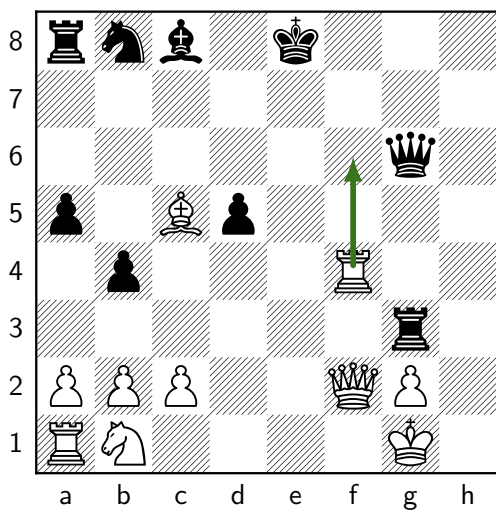


(c)

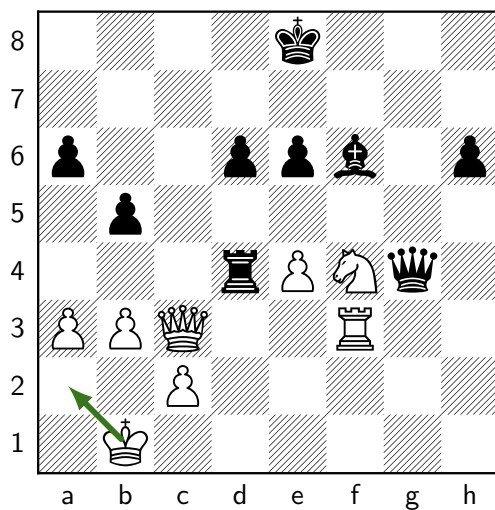


(d)

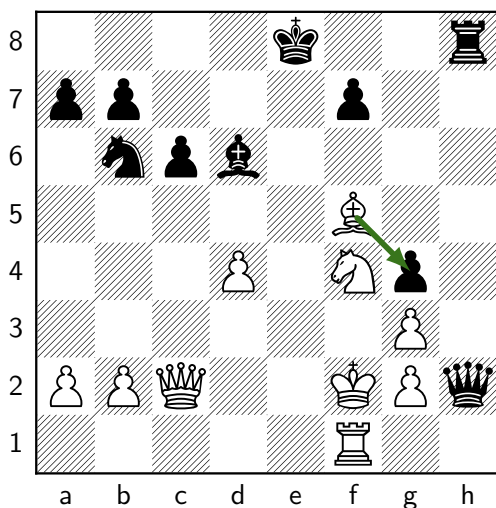
Figure 23: **Move 29, White: Qf2.** (a) Our position; Melanie plays Qf2 (d2f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#604` (ply 72) [structural distance $d=73$, QAP obj=30]: the master played Qh1; under μ our Qd2 plays the role of that game’s Qd1; cross-type roles: our Nb1 as their Ra1, our Pa2 as their Bg2; transferred move d2f2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1344` (ply 69) [structural distance $d=95$, QAP obj=110]: the master played Kg7; under μ our Kg1 plays the role of that game’s Kh1; cross-type roles: our rg3 as their Qe7, our Nb1 as their bg6; transferred move g1f2. (d) Retrieved from `ccr1_top_p0003#2606` (ply 58) [structural distance $d=95$, QAP obj=102]: the master played h4; under μ our Pc2 plays the role of that game’s Ph3; cross-type roles: our Bc5 as their Rd1, our Ra1 as their Pa2; transferred move c2c4. Synthesis: 1 retrieved analog transferred to Qf2 (summed QAP objective 30, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



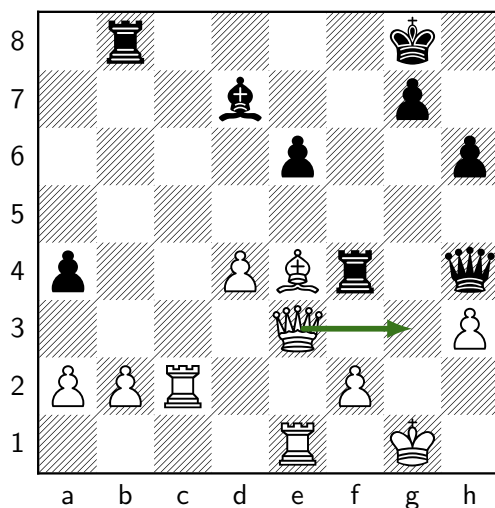
(a)



(b)

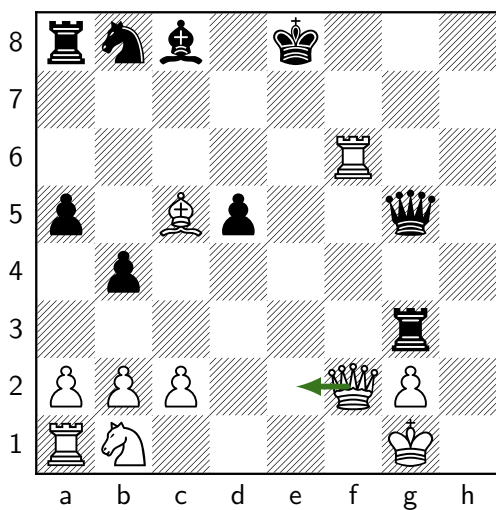


(c)

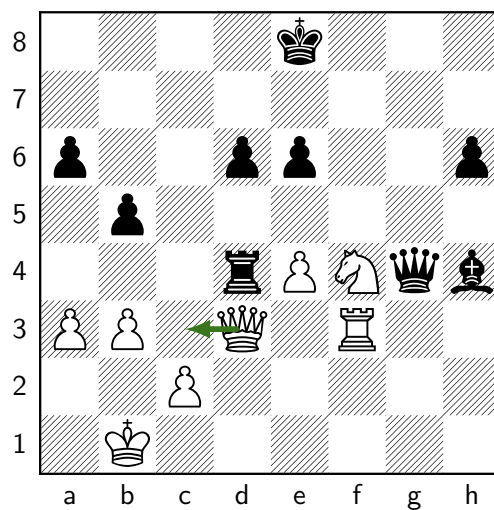


(d)

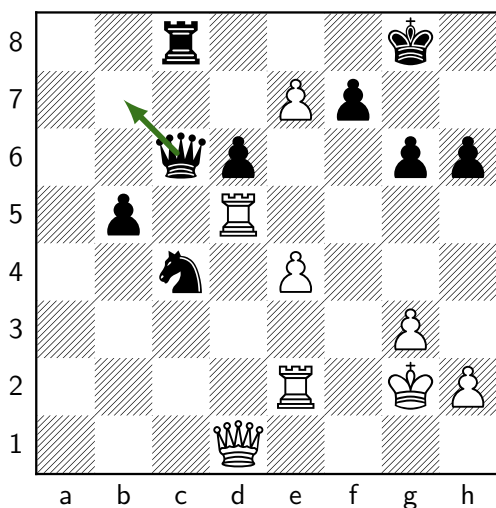
Figure 24: **Move 30, White: Rf6**. (a) Our position; Melanie plays Rf6 (f4f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2312` (ply 72) [structural distance $d=90$, QAP obj=79]: the master played Ka2; under μ our Kg1 plays the role of that game's Kb1; cross-type roles: our Rf4 as their Pc2, our rg3 as their Nf4; transferred move g1h1. (c) Retrieved from `ccr1_top_p0003#269` (ply 52) [structural distance $d=97$, QAP obj=46]: the master played Bxg4; under μ our Bc5 plays the role of that game's Bf5; cross-type roles: our Qf2 as their Rf1, our Ra1 as their Pa2; transferred move c5b4. (d) Retrieved from `ccr1_top_p0001#3650` (ply 60) [structural distance $d=99$, QAP obj=41]: the master played Qg3; under μ our Qf2 plays the role of that game's Qe3; cross-type roles: our Nb1 as their Pa2, our Bc5 as their Pd4; transferred move f2e3. Synthesis: no retrieved analog transferred to Rf6 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



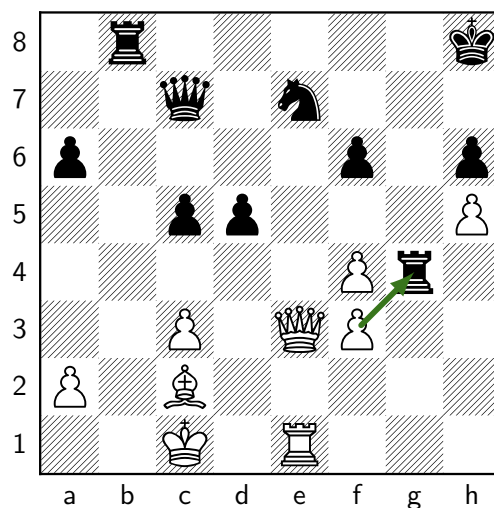
(a)



(b)

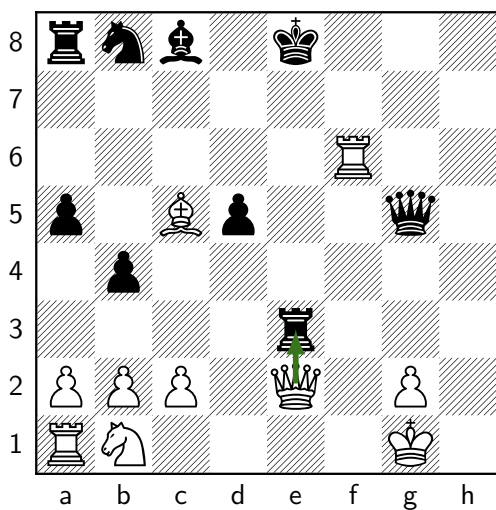


(c)

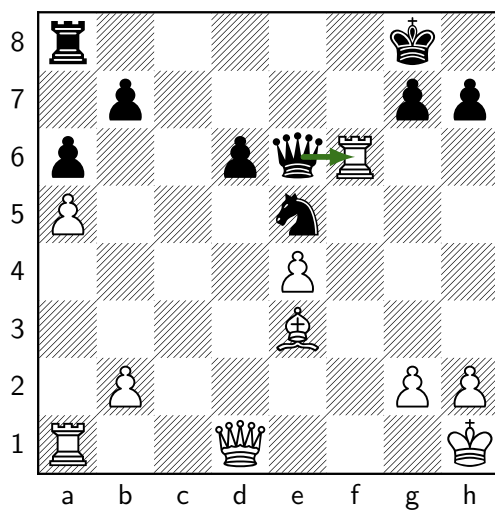


(d)

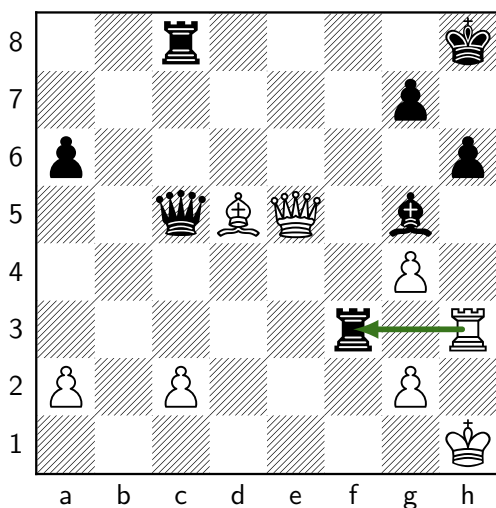
Figure 25: **Move 31, White: Qe2+.** (a) Our position; Melanie plays Qe2+ (f2e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2312` (ply 70) [structural distance $d=84$, QAP obj=71]: the master played Qc3; under μ our Qf2 plays the role of that game's Qd3; cross-type roles: our Bc5 as their Pc2, our Rf6 as their Nf4; transferred move f2e2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2016` (ply 75) [structural distance $d=79$, QAP obj=54]: the master played Qb7; under μ our Qf2 plays the role of that game's Qc3; cross-type roles: our nb8 as their pe7, our Nb1 as their Pf7; transferred move f2e2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#913` (ply 58) [structural distance $d=79$, QAP obj=39]: the master played fxg4; under μ our Qf2 plays the role of that game's Pf3 (a **cross-type** correspondence); cross-type roles: our Pa2 as their Re1, our Ra1 as their Pa2; transferred move f2e2 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Qe2+ (summed QAP objective 246, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



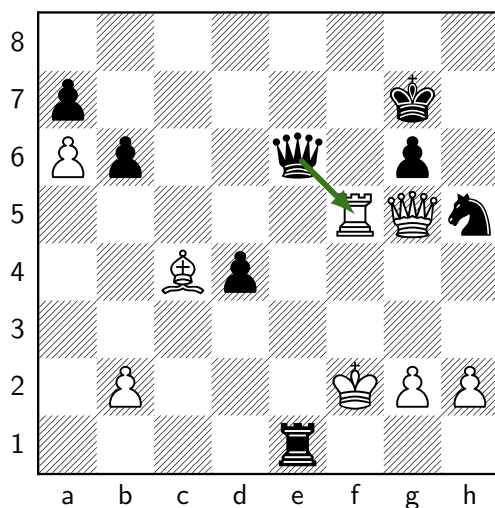
(a)



(b)

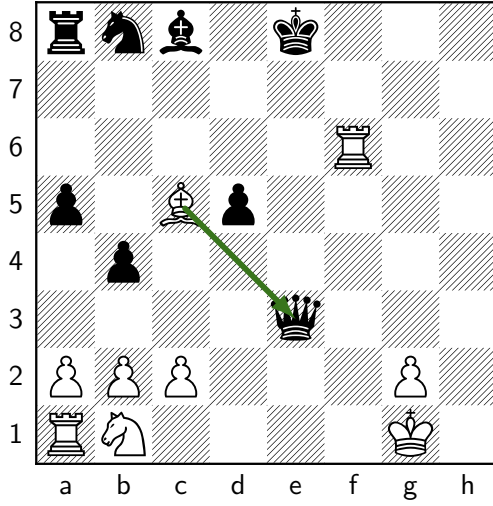


(c)

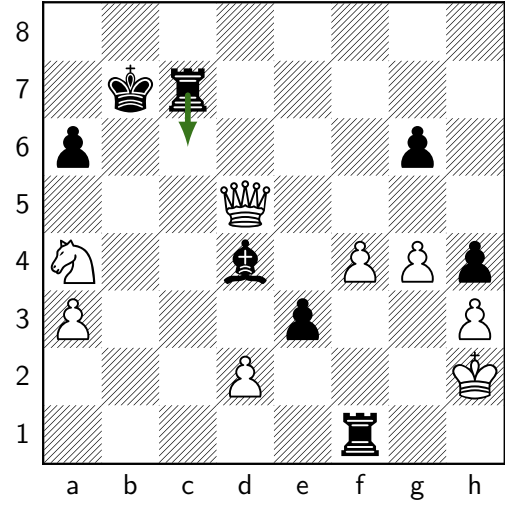


(d)

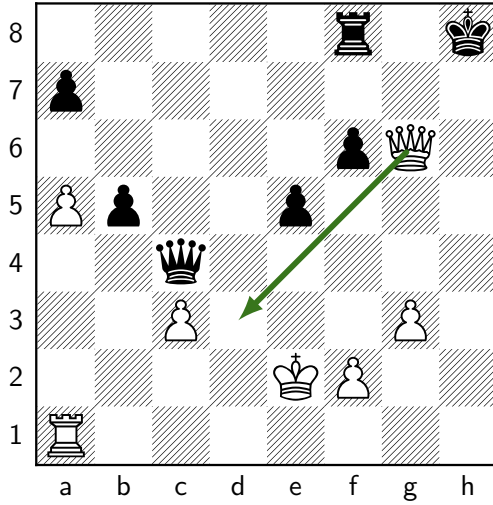
Figure 26: **Move 32, White: Qxe3+.** (a) Our position; Melanie plays Qxe3+ (e2e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2940` (ply 47) [structural distance $d=79$, QAP obj=46]: the master played Qxf6; under μ our Qe2 plays the role of that game’s Qe3; cross-type roles: our pa5 as their Ra8, our nb8 as their Pb7; transferred move e2e3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1721` (ply 64) [structural distance $d=92$, QAP obj=40]: the master played Rxf3; under μ our Qe2 plays the role of that game’s Rh3 (a **cross-type** correspondence); cross-type roles: our Bc5 as their Pg2, our Nb1 as their Pg4; transferred move e2e3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1433` (ply 67) [structural distance $d=91$, QAP obj=35]: the master played Qxf5+; under μ our Qe2 plays the role of that game’s Qe3; cross-type roles: our Nb1 as their Pa7, our Qe2 as their Pg6; transferred move e2e3 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Qxe3+ (summed QAP objective 174, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



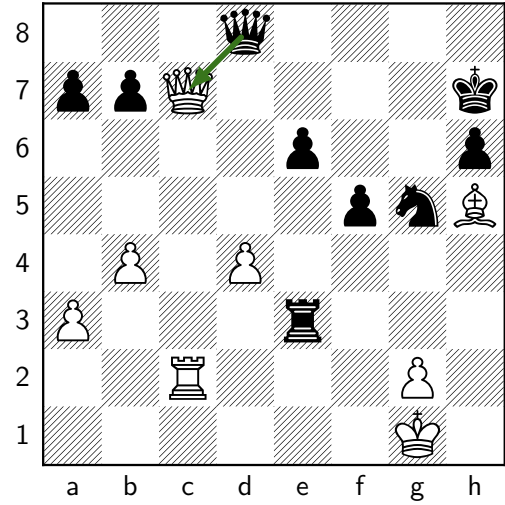
(a)



(b)

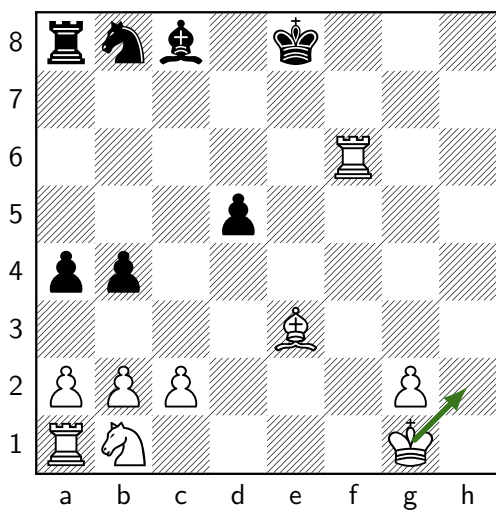


(c)

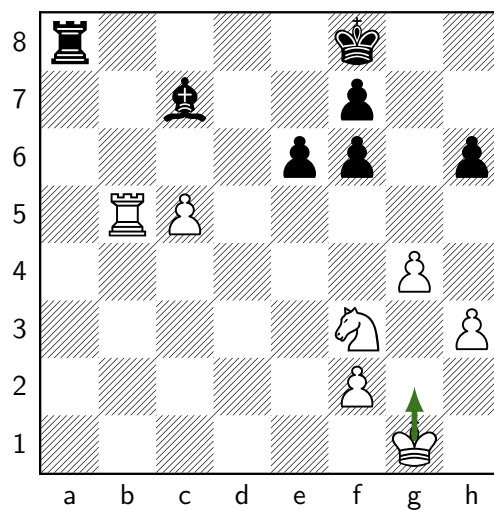


(d)

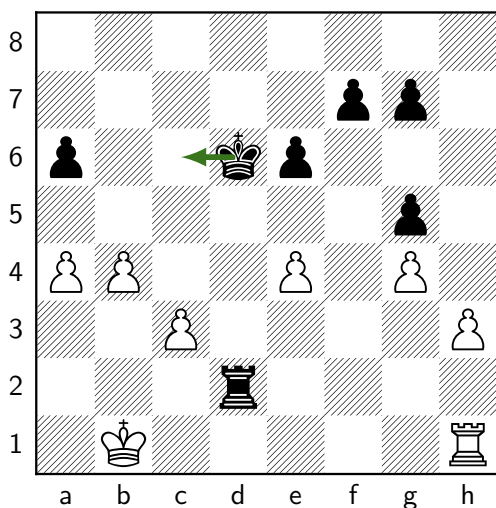
Figure 27: **Move 33, White: Bxe3.** (a) Our position; Melanie plays Bxe3 (c5e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#951` (ply 73) [structural distance $d=69$, QAP obj=68]: the master played Rc6; under μ our Bc5 plays the role of that game’s Rc2 (a **cross-type** correspondence); cross-type roles: our pa5 as their na4, our Ra1 as their pa3; transferred move c5e3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#109` (ply 62) [structural distance $d=81$, QAP obj=68]: the master played Qd3; under μ our Bc5 plays the role of that game’s Qg6 (a **cross-type** correspondence); cross-type roles: our Nb1 as their Pc3, our bc8 as their Pa5; transferred move c5e3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1272` (ply 77) [structural distance $d=85$, QAP obj=34]: the master played Qxc7; under μ our Bc5 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Ra1 as their Pe6, our Nb1 as their Pf5; transferred move c5e3 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Bxe3 (summed QAP objective 266, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



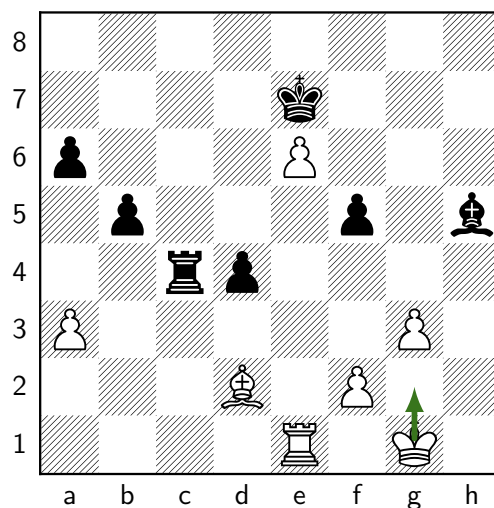
(a)



(b)

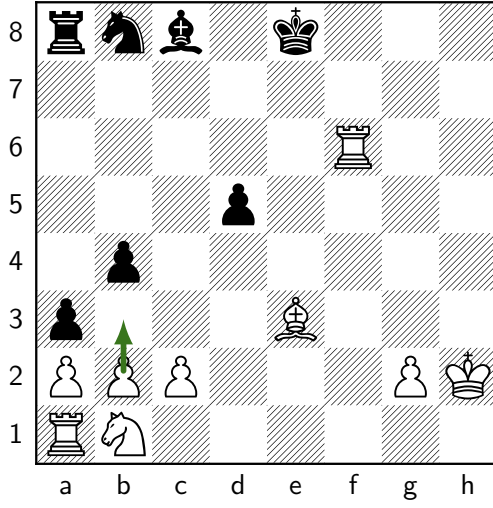


(c)

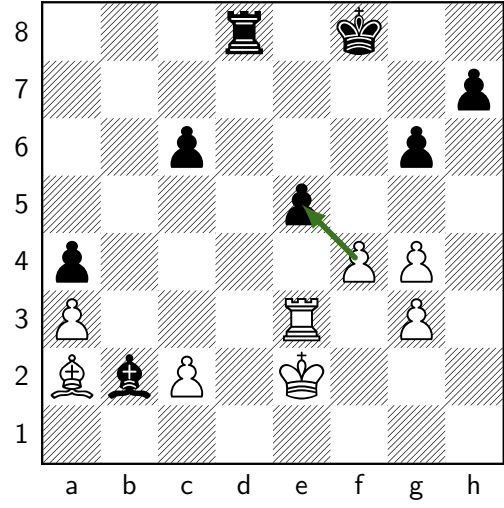


(d)

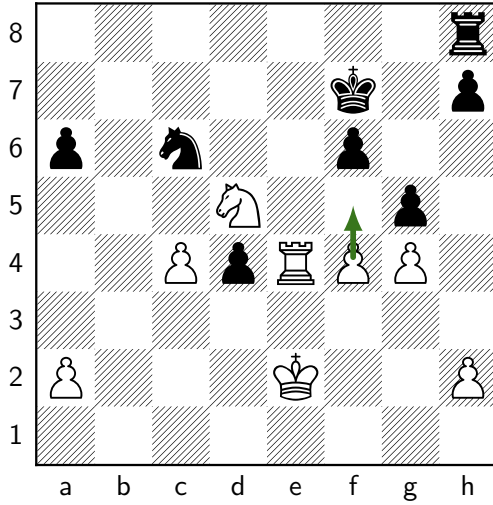
Figure 28: **Move 34, White: Kh2.** (a) Our position; Melanie plays Kh2 (g1h2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2910` (ply 58) [structural distance $d=38$, QAP obj=5]: the master played Kg2; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Be3 as their Nf3, our Nb1 as their Ph3; transferred move g1h2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2571` (ply 57) [structural distance $d=43$, QAP obj=4]: the master played Kc6; under μ our Kg1 plays the role of that game’s Kd3; cross-type roles: our Ra1 as their Pg7, our Rf6 as their Pa6; transferred move g1h2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1609` (ply 78) [structural distance $d=42$, QAP obj=4]: the master played Kg2; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Be3 as their Re1, our pb4 as their Bd2; transferred move g1h2 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Kh2 (summed QAP objective 27, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



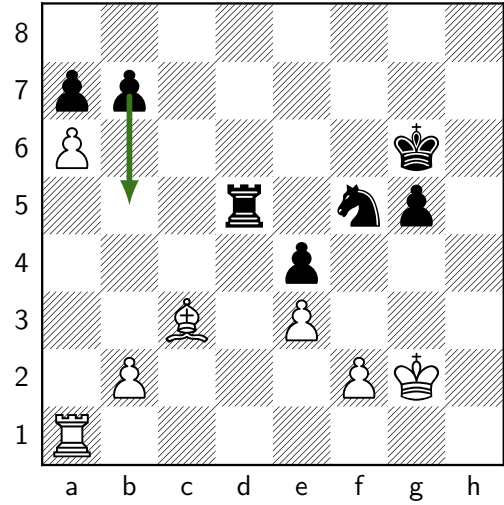
(a)



(b)



(c)



(d)

Figure 29: **Move 35, White: b3.** (a) Our position; Melanie plays b3 (b2b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#570` (ply 66) [structural distance $d=41$, QAP obj=79]: the master played `fxe5`; under μ our *Pb2* plays the role of that game’s *Pf4*; cross-type roles: our *Rf6* as their *Ba2*, our *pa3* as their *bb2*; transferred move `b2b3` (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3661` (ply 60) [structural distance $d=40$, QAP obj=53]: the master played `f5`; under μ our *Pb2* plays the role of that game’s *Pf4*; cross-type roles: our *Ra1* as their *Pa2*, our *bc8* as their *Pc4*; transferred move `b2b3` (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2462` (ply 71) [structural distance $d=40$, QAP obj=51]: the master played `b5`; under μ our *Pb2* plays the role of that game’s *Pb2*; cross-type roles: our *Rf6* as their *pb2*; transferred move `b2b3` (**endorses this move**). Synthesis: 10 retrieved analogs transferred to b3 (summed QAP objective 413, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

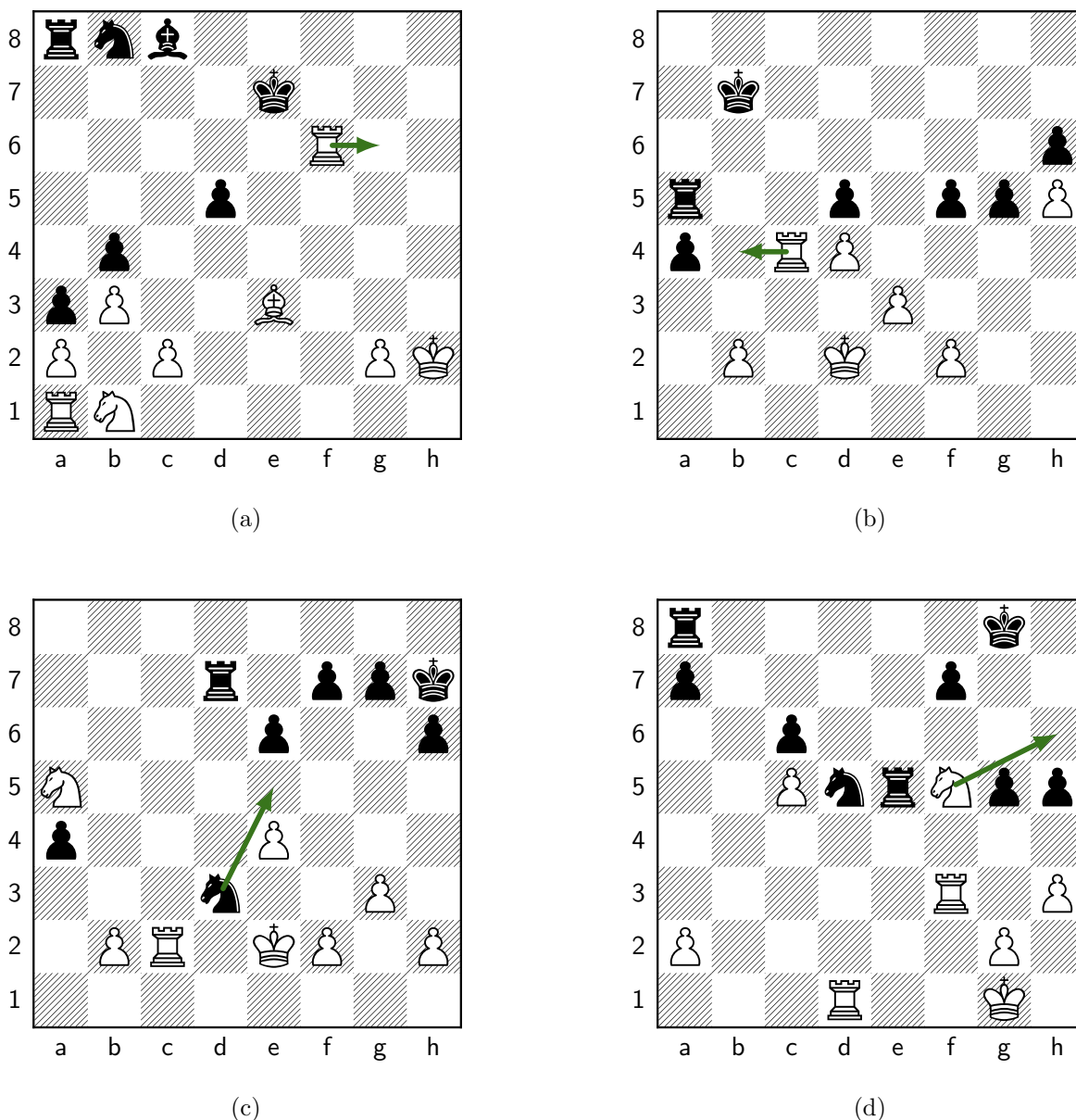
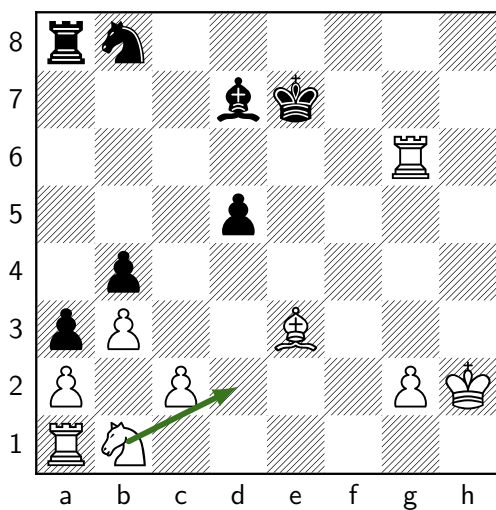
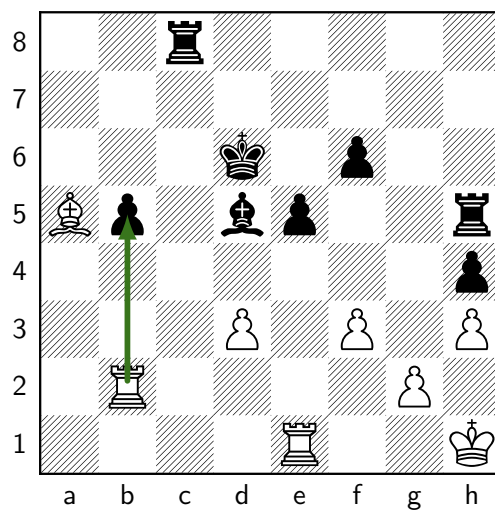


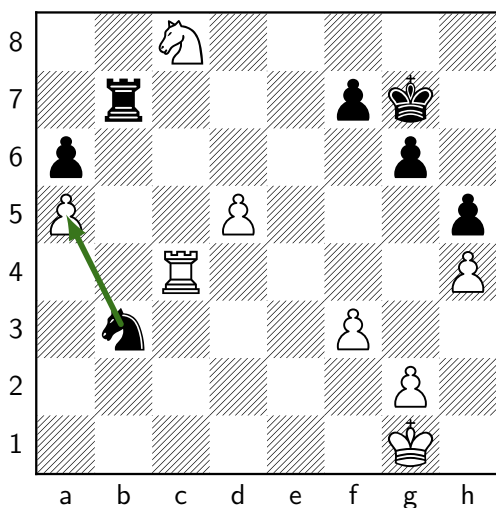
Figure 30: **Move 36, White: Rg6.** (a) Our position; Melanie plays Rg6 (f6g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#1999 (ply 86) [structural distance $d=34$, QAP obj=18]: the master played Rb4+; under μ our Rf6 plays the role of that game’s Rc4; cross-type roles: our Nb1 as their pa4, our nb8 as their Pd4; transferred move f6g6 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#3281 (ply 67) [structural distance $d=39$, QAP obj=14]: the master played Ne5; under μ our Rf6 plays the role of that game’s Nd6 (a **cross-type** correspondence); cross-type roles: our Pb3 as their Rd7, our Ra1 as their Pf7; transferred move f6g6 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#3632 (ply 64) [structural distance $d=39$, QAP obj=9]: the master played Nh6+; under μ our Rf6 plays the role of that game’s Nf5 (a **cross-type** correspondence); cross-type roles: our Pg2 as their Rd1, our Ra1 as their Pa2; transferred move f6g6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rg6 (summed QAP objective 48, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



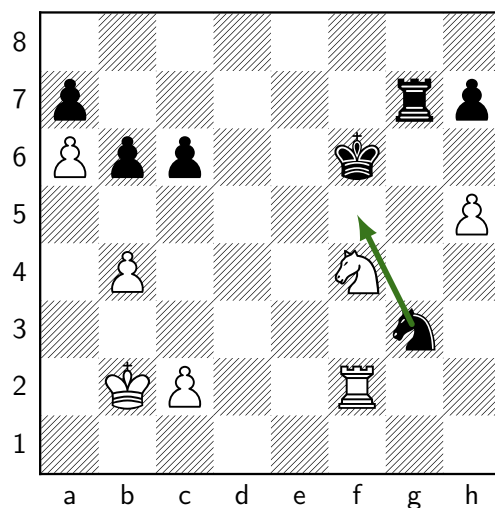
(a)



(b)

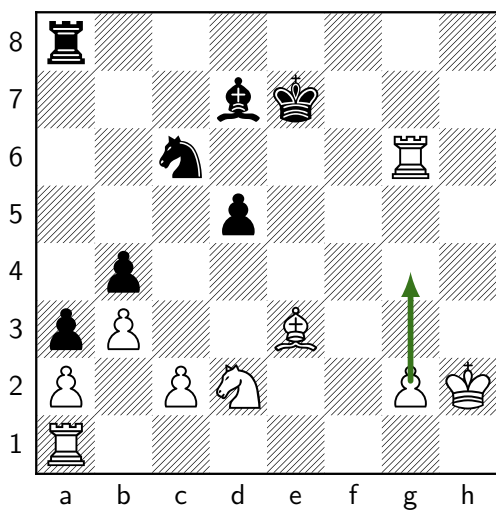


(c)

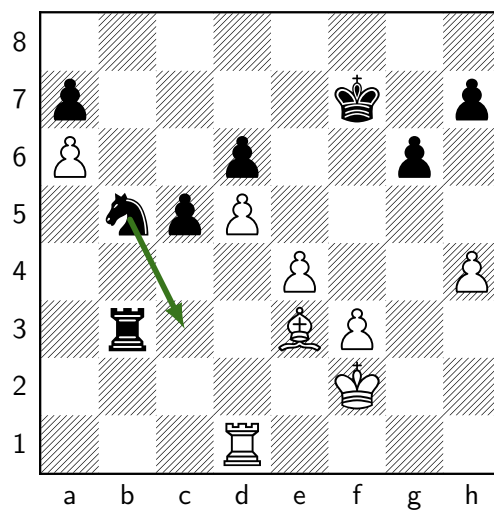


(d)

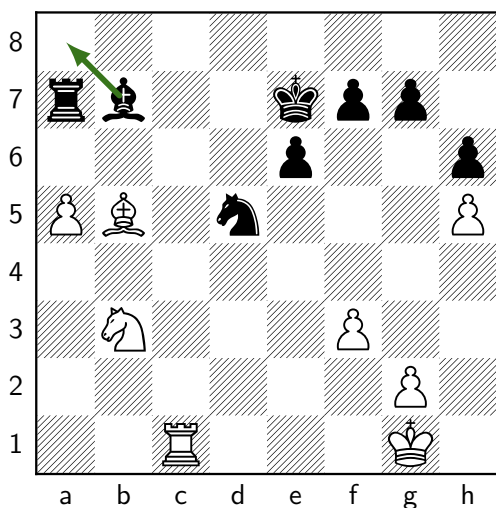
Figure 31: **Move 37, White: Nd2.** (a) Our position; Melanie plays Nd2 (b1d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2174` (ply 68) [structural distance $d=46$, QAP obj=24]: the master played $Rxb5$; under μ our $Nb1$ plays the role of that game’s $Rb2$ (a **cross-type** correspondence); cross-type roles: our $ra8$ as their $Pd3$, our $nb8$ as their $Ba5$; transferred move b1d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#360` (ply 87) [structural distance $d=31$, QAP obj=24]: the master played $Nxa5$; under μ our $Nb1$ plays the role of that game’s $Nb6$; cross-type roles: our $Be3$ as their $ph4$, our $bd7$ as their $pg2$; transferred move b1d2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1180` (ply 89) [structural distance $d=41$, QAP obj=22]: the master played $Nf5$; under μ our $Nb1$ plays the role of that game’s $Ng6$; cross-type roles: our $bd7$ as their $Pb6$, our $Rg6$ as their $pb4$; transferred move b1d2 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Nd2 (summed QAP objective 172, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



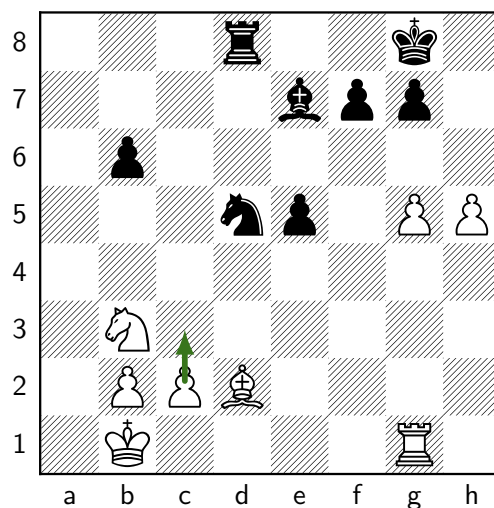
(a)



(b)

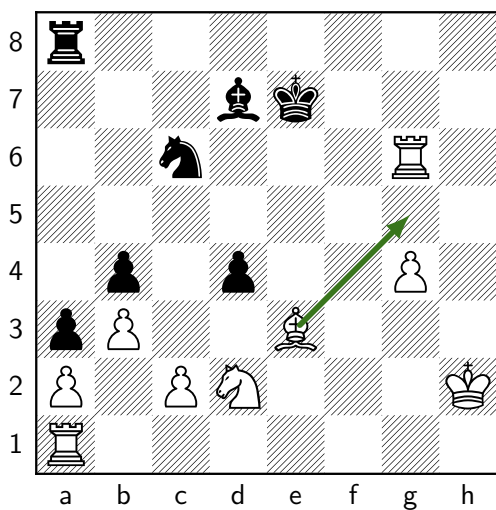


(c)

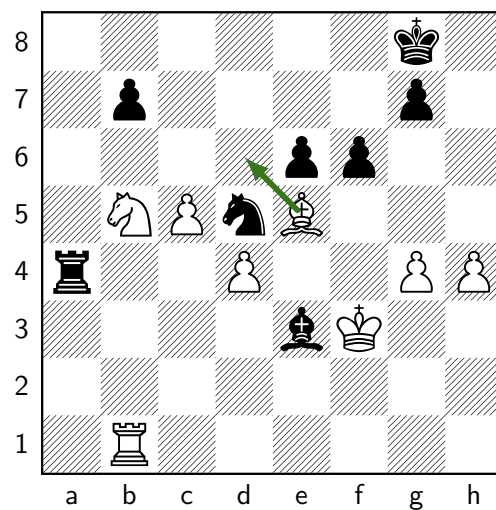


(d)

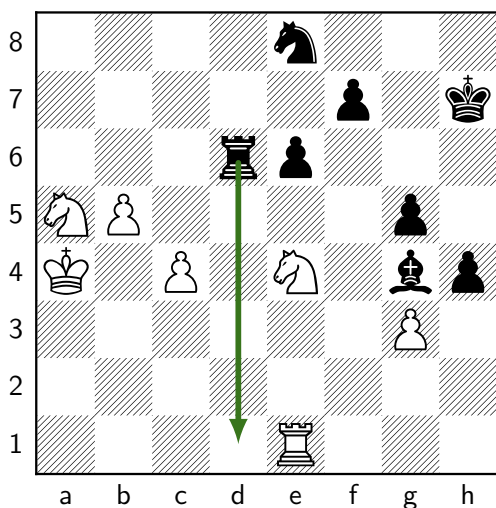
Figure 32: **Move 38, White: g4.** (a) Our position; Melanie plays g4 (g2g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3530` (ply 87) [structural distance $d=40$, QAP obj=20]: the master played Nc3; under μ our Pg2 plays the role of that game’s Nb4 (a **cross-type** correspondence); cross-type roles: our Be3 as their Pc5, our Nd2 as their pe4; transferred move g2g4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#651` (ply 77) [structural distance $d=39$, QAP obj=16]: the master played Ba8; under μ our Pg2 plays the role of that game’s Bb2 (a **cross-type** correspondence); cross-type roles: our Ra1 as their Pg7, our ra8 as their Ph6; transferred move g2g4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2736` (ply 60) [structural distance $d=34$, QAP obj=16]: the master played c3; under μ our Pg2 plays the role of that game’s Pc2; cross-type roles: our bd7 as their Rg1, our Pb3 as their Bd2; transferred move g2g4 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to g4 (summed QAP objective 92, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



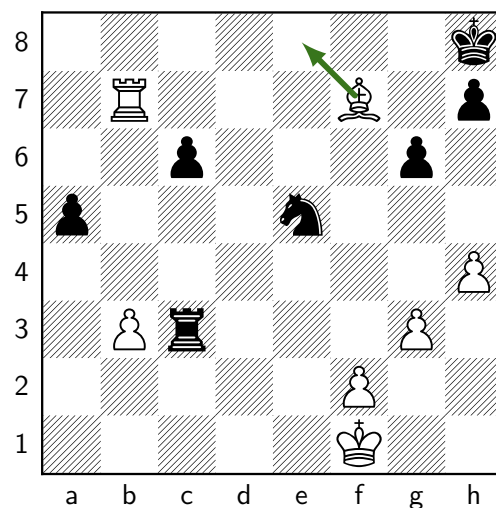
(a)



(b)

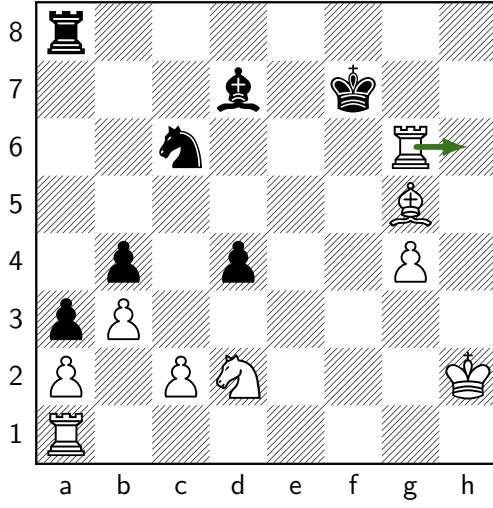


(c)

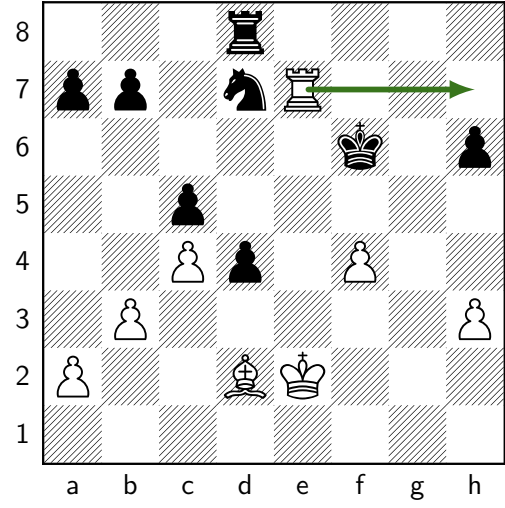


(d)

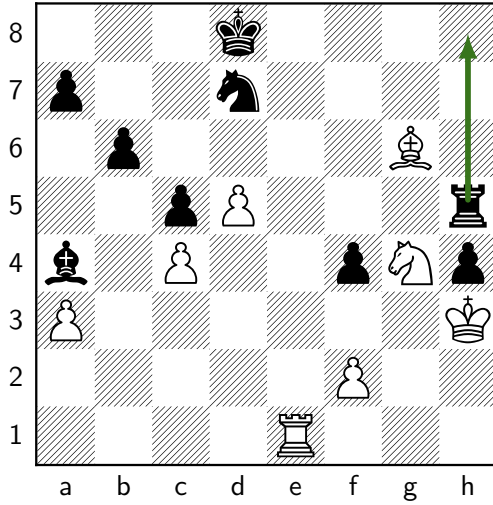
Figure 33: **Move 39, White: Bg5+.** (a) Our position; Melanie plays Bg5+ (e3g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1776` (ply 72) [structural distance $d=38$, QAP obj=33]: the master played Bd6; under μ our Be3 plays the role of that game’s Be5; cross-type roles: our Rg6 as their be3, our nc6 as their Pd4; transferred move e3g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3776` (ply 79) [structural distance $d=46$, QAP obj=22]: the master played Rd1; under μ our Be3 plays the role of that game’s Rd3 (a **cross-type** correspondence); cross-type roles: our Pc2 as their Ne8, our Pb3 as their na5; transferred move e3g5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#513` (ply 84) [structural distance $d=45$, QAP obj=17]: the master played Be8; under μ our Be3 plays the role of that game’s Bf7; cross-type roles: our nc6 as their Pb3, our Ra1 as their Ph4; transferred move e3g5 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Bg5+ (summed QAP objective 154, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



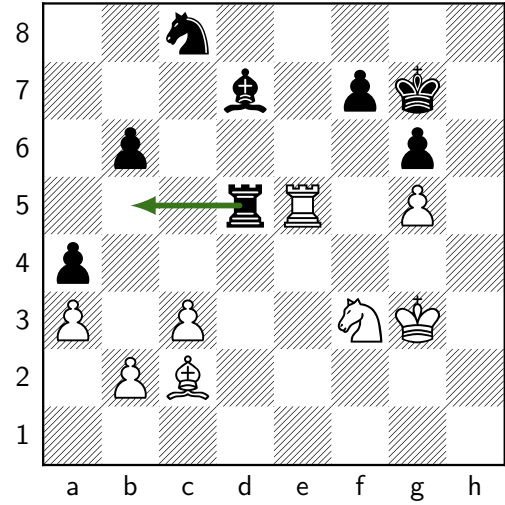
(a)



(b)

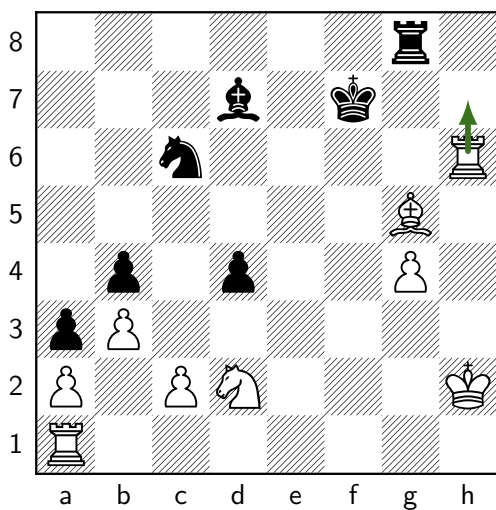


(c)

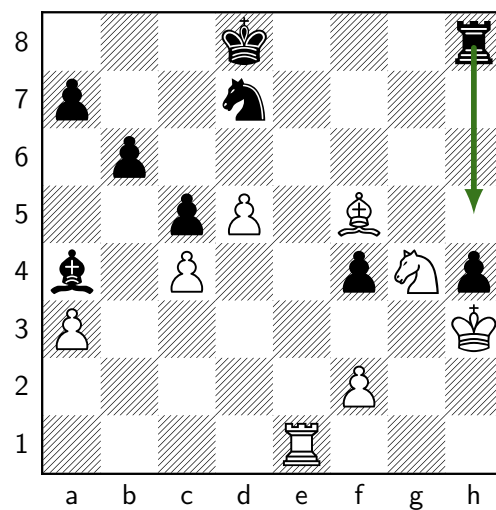


(d)

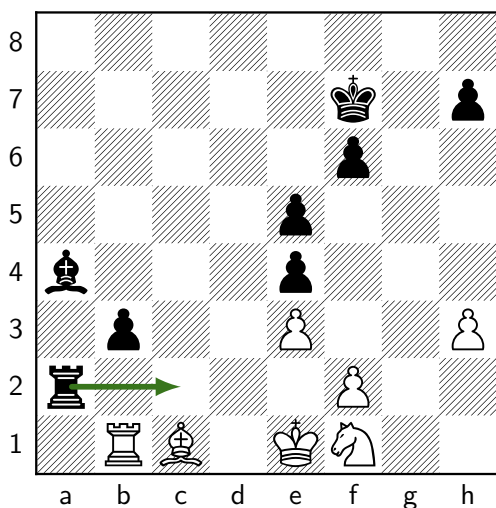
Figure 34: **Move 40, White: Rh6.** (a) Our position; Melanie plays Rh6 (**g6h6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1345` (ply 70) [structural distance $d=45$, QAP obj=32]: the master played Rh7; under μ our Rg6 plays the role of that game’s Re7; cross-type roles: our Pc2 as their Bd2, our Ra1 as their Ph3; transferred move g6h6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#3068` (ply 81) [structural distance $d=43$, QAP obj=23]: the master played Rh8; under μ our Rg6 plays the role of that game’s Rh4; cross-type roles: our Ra1 as their Pa7, our kf7 as their bg6; transferred move g6h6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#131` (ply 79) [structural distance $d=36$, QAP obj=21]: the master played Rb5; under μ our Rg6 plays the role of that game’s Rd4; cross-type roles: our kf7 as their re5, our ra8 as their pc3; transferred move g6h6 (**endorses this move**). Synthesis: 16 retrieved analogs transferred to Rh6 (summed QAP objective 250, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



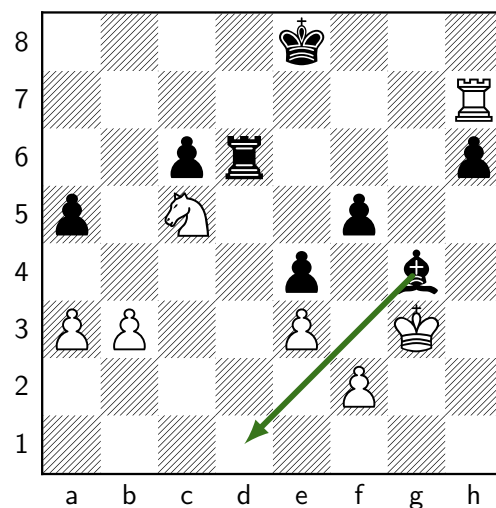
(a)



(b)

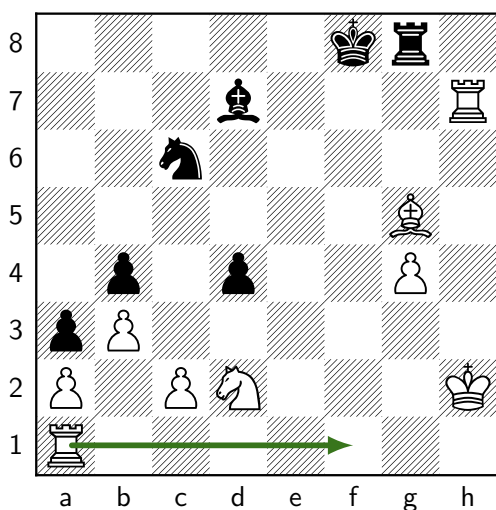


(c)

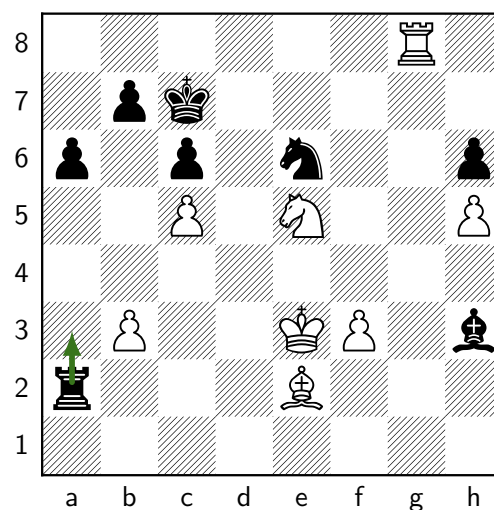


(d)

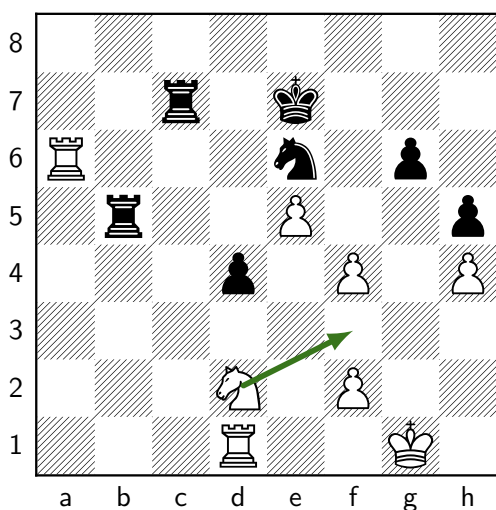
Figure 35: **Move 41, White: Rh7+.** (a) Our position; Melanie plays Rh7+ (h6h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3068` (ply 79) [structural distance $d=38$, QAP obj=16]: the master played Rh5; under μ our Rh6 plays the role of that game’s Rh1; cross-type roles: our Bg5 as their Nd7, our Ra1 as their pd5; transferred move h6h7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#488` (ply 75) [structural distance $d=41$, QAP obj=13]: the master played Rc2; under μ our Rh6 plays the role of that game’s Ra7; cross-type roles: our Nd2 as their Pf6, our Pg4 as their Ba4; transferred move h6h7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3560` (ply 95) [structural distance $d=43$, QAP obj=10]: the master played Bd1; under μ our Rh6 plays the role of that game’s Bg5 (a **cross-type** correspondence); cross-type roles: our Ra1 as their Pc6, our Pg4 as their Rd6; transferred move h6h7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rh7+ (summed QAP objective 55, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



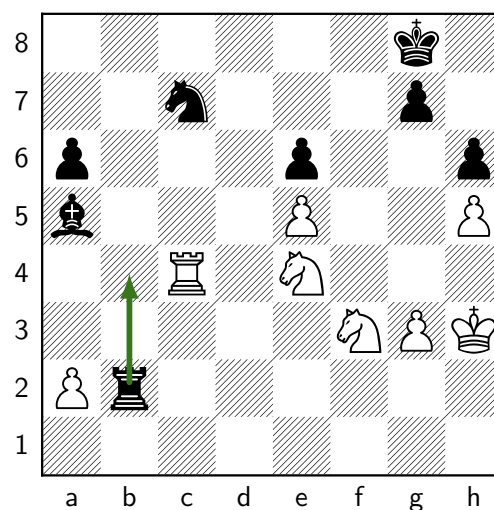
(a)



(b)

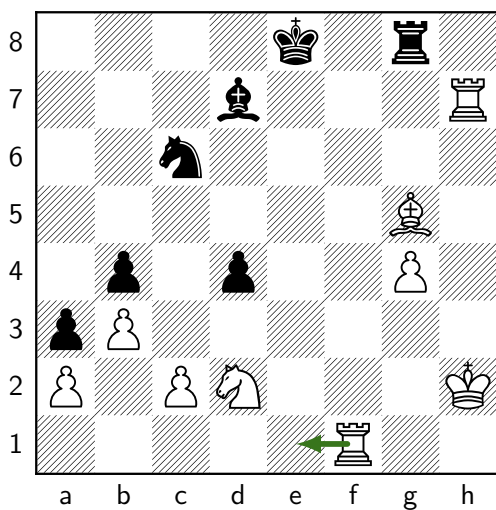


(c)

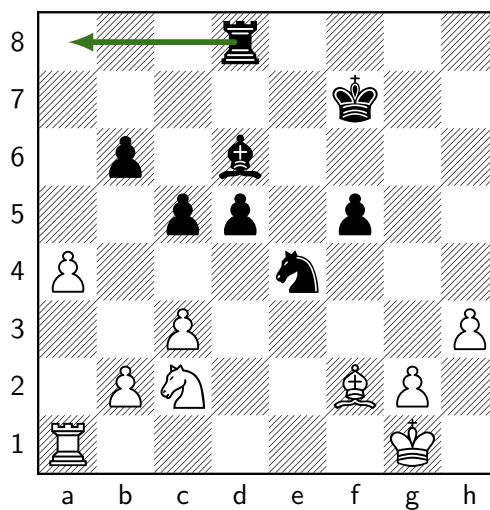


(d)

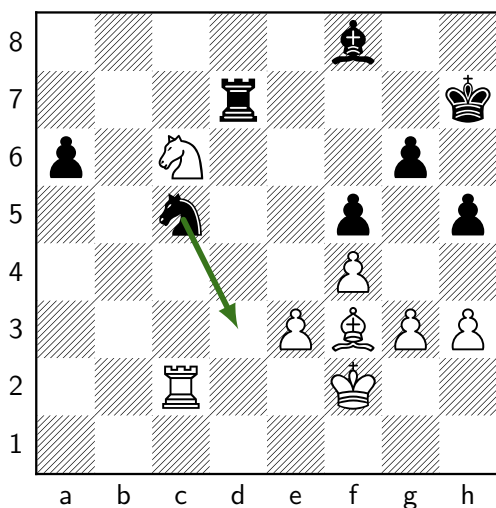
Figure 36: **Move 42, White: Rf1+.** (a) Our position; Melanie plays Rf1+ (a1f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2691` (ply 79) [structural distance $d=42$, QAP obj=43]: the master played Ra3; under μ our Rh7 plays the role of that game's Ra7; cross-type roles: our Pb3 as their rg8, our Bg5 as their Pc6; transferred move h7d7. (c) Retrieved from `ccr1_top_p0003#1019` (ply 82) [structural distance $d=37$, QAP obj=39]: the master played Nf3; under μ our Nd2 plays the role of that game's Nd2; cross-type roles: our Bg5 as their Pe5, our bd7 as their ne6; transferred move d2f3. (d) Retrieved from `ccr1_top_p0000#1319` (ply 69) [structural distance $d=42$, QAP obj=36]: the master played Rb4; under μ our Rh7 plays the role of that game's Rb7; cross-type roles: our Bg5 as their Nc7, our nc6 as their Pg7; transferred move h7h6. Synthesis: no retrieved analog transferred to Rf1+ – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



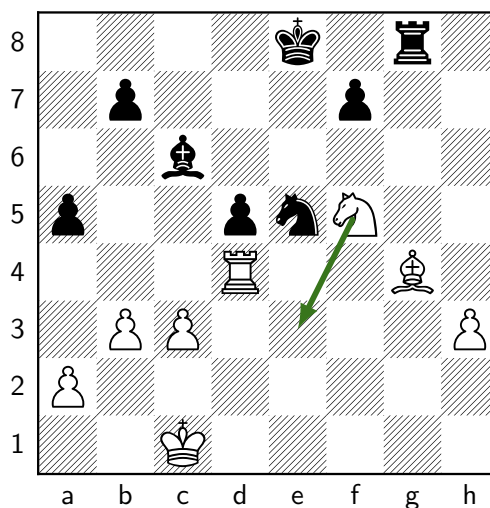
(a)



(b)

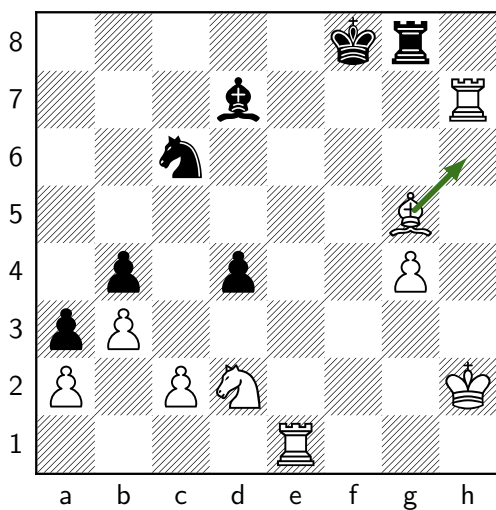


(c)

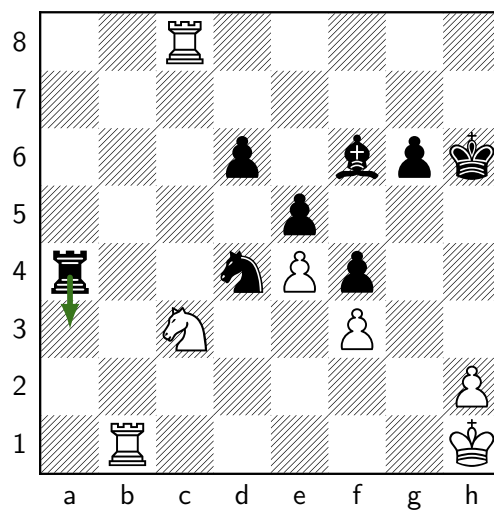


(d)

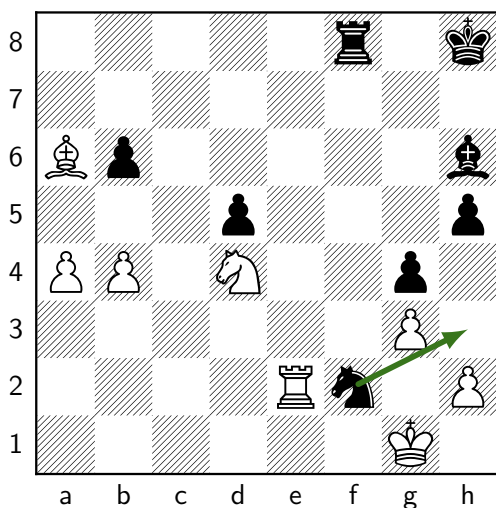
Figure 37: **Move 43, White: Re1+ (f1e1).** (a) Our position; Melanie plays Re1+ (f1e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1607` (ply 73) [structural distance $d=52$, QAP obj=34]: the master played Ra8; under μ our Rf1 plays the role of that game’s Rd1; cross-type roles: our Pb3 as their Bd6, our nc6 as their Pd5; transferred move f1e1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1010` (ply 73) [structural distance $d=51$, QAP obj=32]: the master played Nd3+; under μ our Bg5 plays the role of that game’s Nc4 (a **cross-type** correspondence); cross-type roles: our Pg4 as their Bf8, our pa3 as their nc6; transferred move g5f6. (d) Retrieved from `ccr1_top_p0000#869` (ply 70) [structural distance $d=42$, QAP obj=30]: the master played Ne3; under μ our Nd2 plays the role of that game’s Nf5; cross-type roles: our bd7 as their pd5, our rg8 as their ne5; transferred move d2e4. Synthesis: 1 retrieved analog transferred to Re1+ (summed QAP objective 34, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



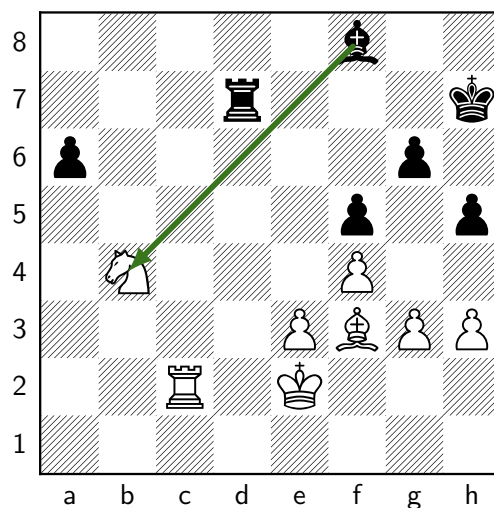
(a)



(b)

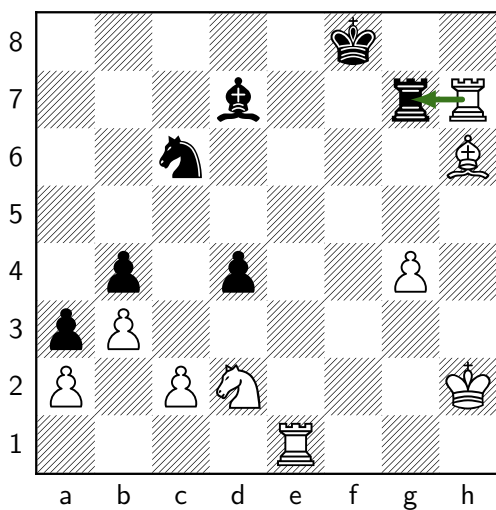


(c)

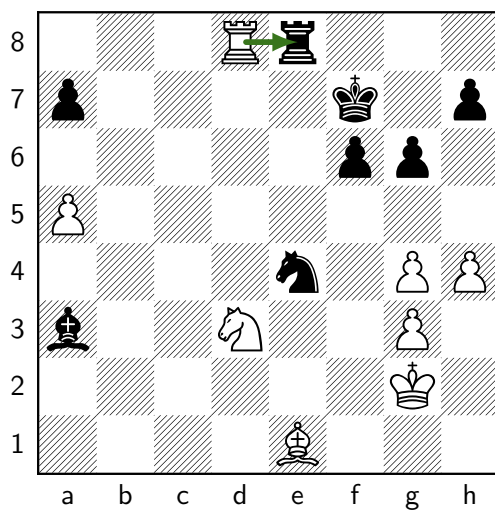


(d)

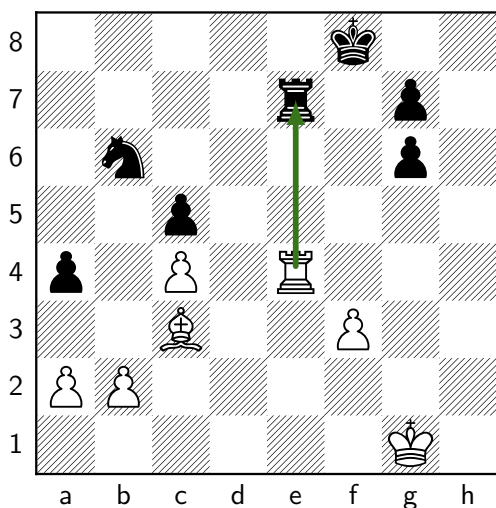
Figure 38: **Move 44, White: Bh6+**. (a) Our position; Melanie plays Bh6+ (g5h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#397` (ply 85) [structural distance $d=40$, QAP obj=11]: the master played Ra3; under μ our Bg5 plays the role of that game’s Ra5 (a **cross-type** correspondence); cross-type roles: our bd7 as their rc8, our Re1 as their Bf6; transferred move g5h6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2421` (ply 75) [structural distance $d=40$, QAP obj=10]: the master played Nh3+; under μ our Bg5 plays the role of that game’s Nf7 (a **cross-type** correspondence); cross-type roles: our pd4 as their ba6, our Rh7 as their Bh6; transferred move g5h6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1010` (ply 77) [structural distance $d=42$, QAP obj=47]: the master played Bxb4; under μ our Rh7 plays the role of that game’s Bf1 (a **cross-type** correspondence); cross-type roles: our Bg5 as their Ph5, our bd7 as their nb4; transferred move h7d7. Synthesis: 2 retrieved analogs transferred to Bh6+ (summed QAP objective 21, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



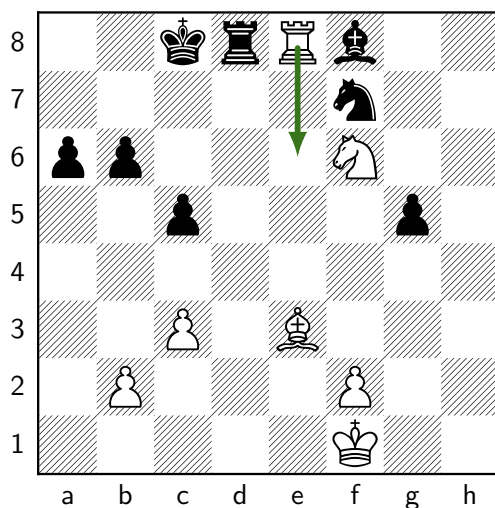
(a)



(b)



(c)



(d)

Figure 39: **Move 45, White: R_xg7.** (a) Our position; Melanie plays R_xg7 (h7g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#2494 (ply 80) [structural distance $d=48$, QAP obj=44]: the master played R_xe8; under μ our R_h7 plays the role of that game’s R_d8; cross-type roles: our P_g4 as their B_e1, our R_e1 as their N_d3; transferred move h7g7 (**endorses this move**). (c) Retrieved from ccr1_top_p0003#522 (ply 78) [structural distance $d=44$, QAP obj=19]: the master played R_xe7; under μ our R_h7 plays the role of that game’s R_e4; cross-type roles: our p_b4 as their n_b6, our n_c6 as their p_g6; transferred move h7g7 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#1774 (ply 72) [structural distance $d=48$, QAP obj=55]: the master played R_e6; under μ our B_h6 plays the role of that game’s R_e8 (a **cross-type** correspondence); cross-type roles: our R_e1 as their P_b2, our N_d2 as their B_e3; transferred move h6f4. Synthesis: 2 retrieved analogs transferred to R_xg7 (summed QAP objective 63, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

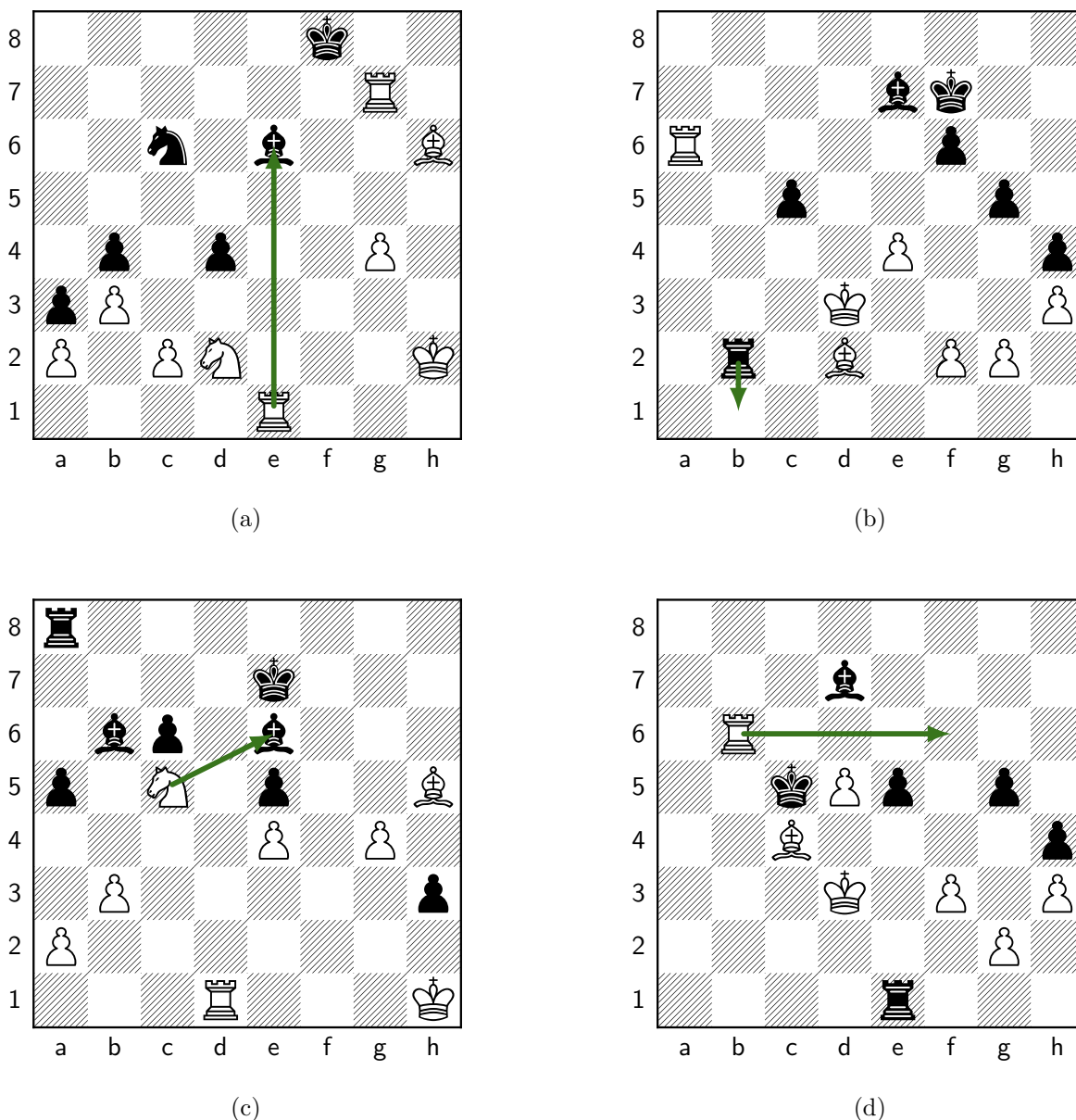
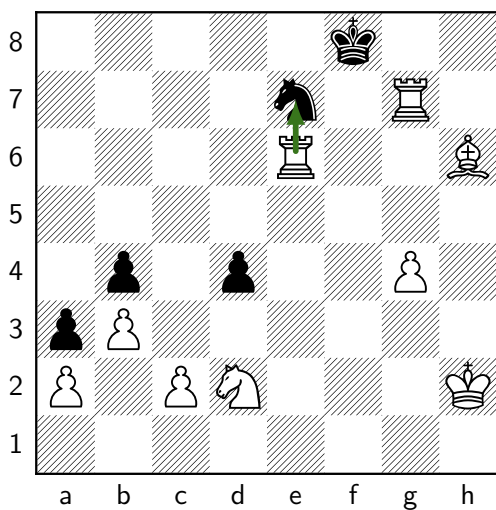
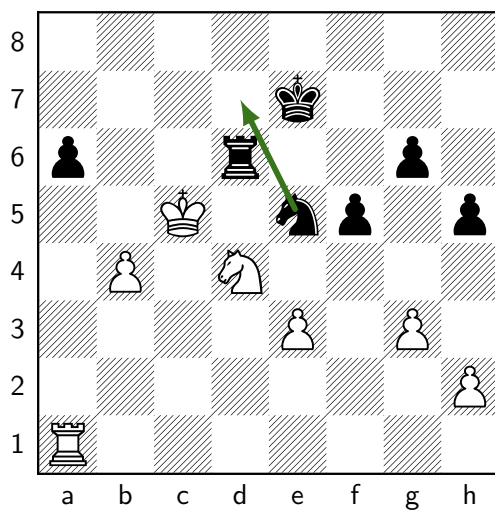


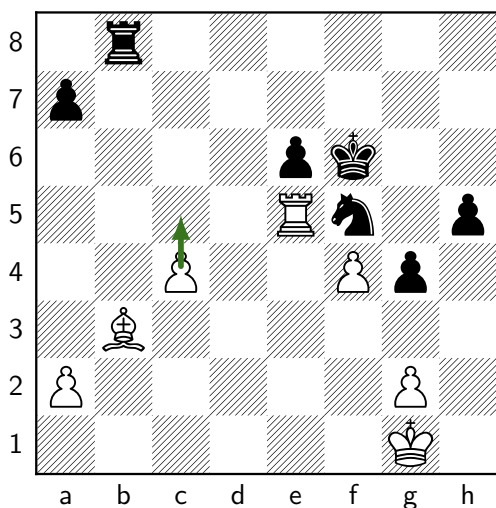
Figure 40: **Move 46, White: Rxe6.** (a) Our position; Melanie plays Rxe6 (e1e6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1869` (ply 91) [structural distance $d=50$, QAP obj=13]: the master played Rb1; under μ our *Re1* plays the role of that game’s *Rb7*; cross-type roles: our *Rg7* as their *Be7*, our *be6* as their *ra6*; transferred move e1e6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3194` (ply 74) [structural distance $d=49$, QAP obj=11]: the master played Nx6; under μ our *Re1* plays the role of that game’s *Nc5* (a **cross-type** correspondence); cross-type roles: our *be6* as their *ph3*, our *Nd2* as their *pa5*; transferred move e1e6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#645` (ply 104) [structural distance $d=50$, QAP obj=29]: the master played Rf6; under μ our *Rg7* plays the role of that game’s *Rb6*; cross-type roles: our *Nd2* as their *re1*, our *Rg7* as their *Bc4*; transferred move g7h7. Synthesis: 2 retrieved analogs transferred to Rxe6 (summed QAP objective 23, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



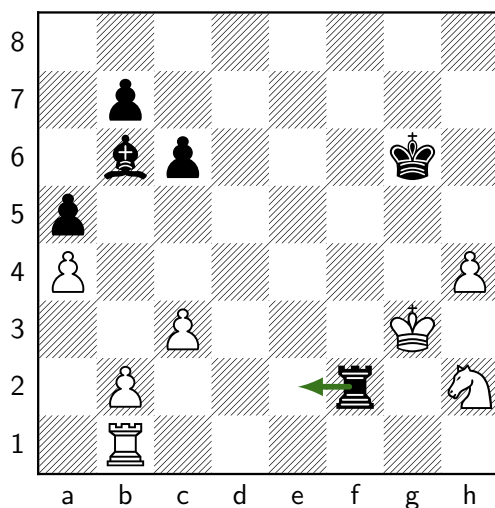
(a)



(b)

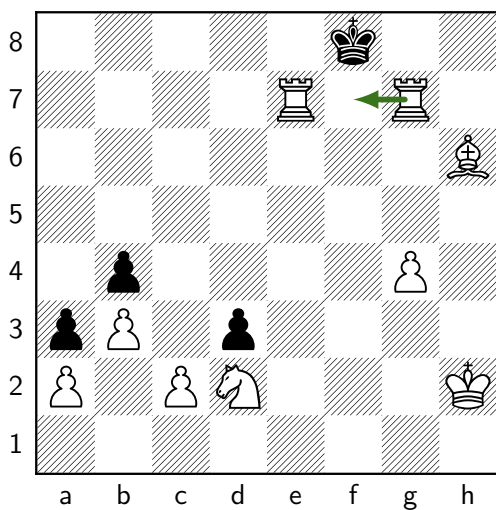


(c)

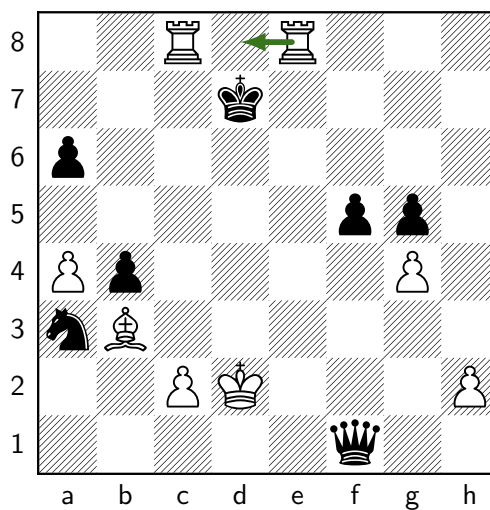


(d)

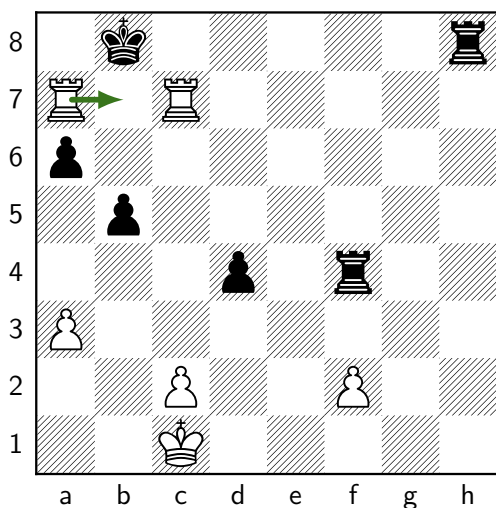
Figure 41: **Move 47, White: Rxe7**. (a) Our position; Melanie plays Rxe7 (e6e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2821` (ply 81) [structural distance $d=44$, QAP obj=46]: the master played Nd7+; under μ our Re6 plays the role of that game’s Ne4 (a **cross-type** correspondence); cross-type roles: our Bh6 as their Pa6, our Nd2 as their ra1; transferred move e6e2. (c) Retrieved from `ccr1_top_p0002#427` (ply 78) [structural distance $d=37$, QAP obj=46]: the master played c5; under μ our Re6 plays the role of that game’s Pc4 (a **cross-type** correspondence); cross-type roles: our pd4 as their Bb3, our Bh6 as their Pf4; transferred move e6e5. (d) Retrieved from `ccr1_top_p0000#216` (ply 79) [structural distance $d=46$, QAP obj=46]: the master played Re2; under μ our Rg7 plays the role of that game’s Rf7; cross-type roles: our Nd2 as their ph4, our Pb3 as their rb1; transferred move g7h7. Synthesis: no retrieved analog transferred to Rxe7 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



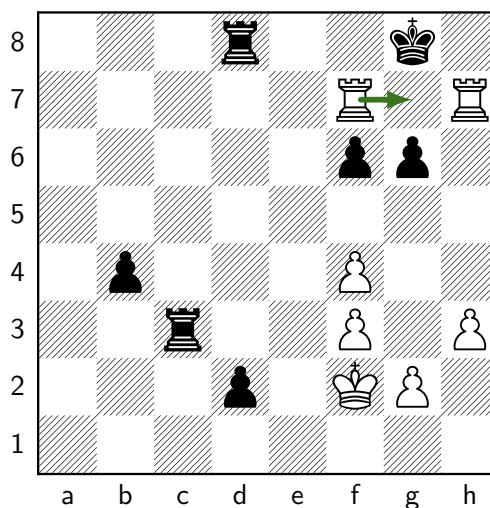
(a)



(b)



(c)



(d)

Figure 42: **Move 48, White: Rgf7+.** (a) Our position; Melanie plays Rgf7+ (g7f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2994` (ply 78) [structural distance $d=54$, QAP obj=55]: the master played Red8+; under μ our Rg7 plays the role of that game’s Re8; cross-type roles: our Pb3 as their qf1, our pd3 as their na3; transferred move g7f7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3366` (ply 90) [structural distance $d=57$, QAP obj=54]: the master played Rab7+; under μ our Rg7 plays the role of that game’s Ra7; cross-type roles: our Nd2 as their rf4, our Pg4 as their rh8; transferred move g7f7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3362` (ply 110) [structural distance $d=55$, QAP obj=53]: the master played Rfg7+; under μ our Rg7 plays the role of that game’s Rf7; cross-type roles: our Nd2 as their rc3, our Bh6 as their Pf3; transferred move g7f7 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Rgf7+ (summed QAP objective 228, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.

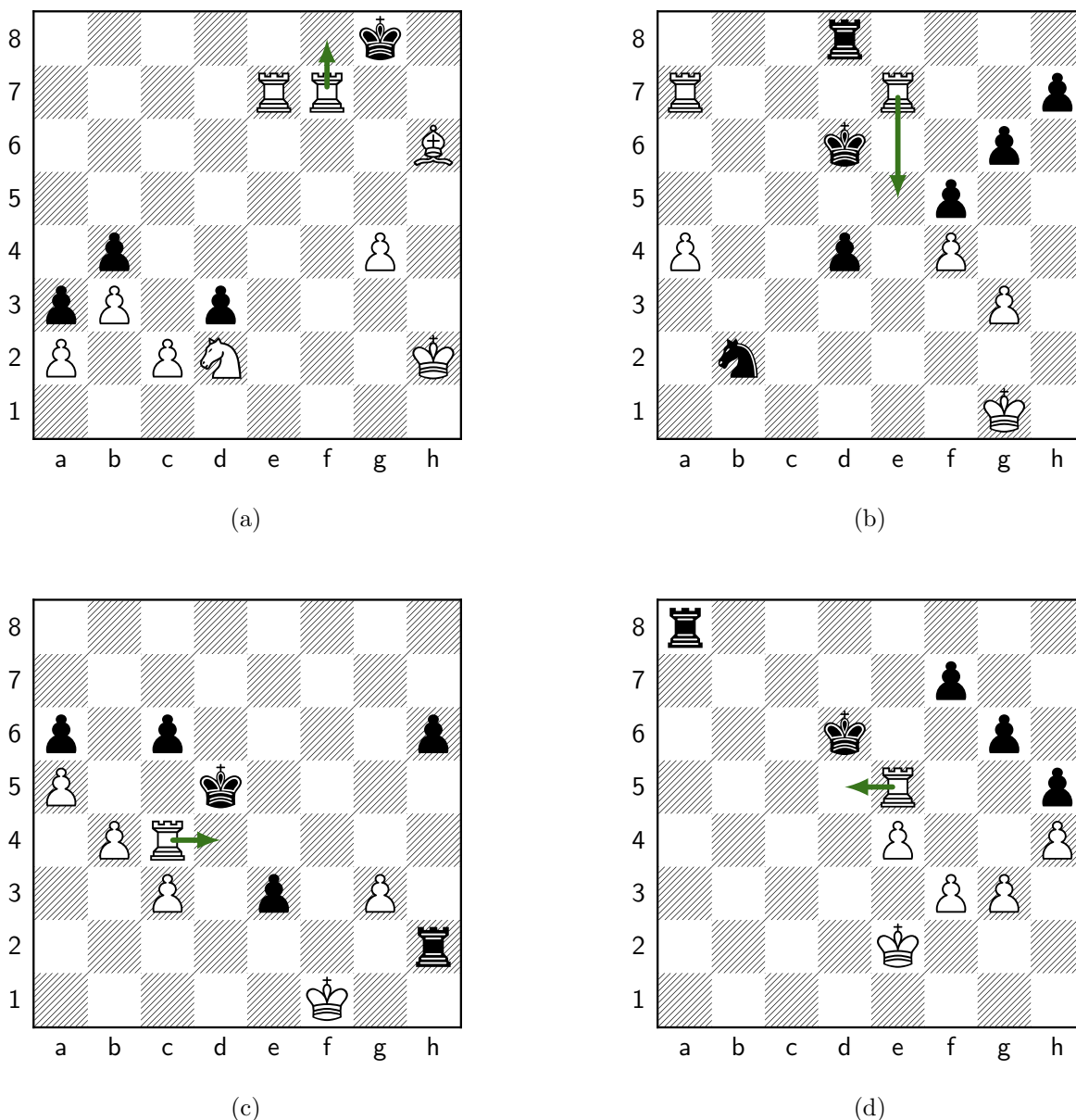


Figure 43: **Move 49, White: Rf8#.** (a) Our position; Melanie plays Rf8# (f7f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#2657 (ply 100) [structural distance $d=33$, QAP obj=23]: the master played Re5; under μ our Rf7 plays the role of that game’s Re7; cross-type roles: our pd3 as their nb2, our Nd2 as their Pa4; transferred move f7f8 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#491 (ply 100) [structural distance $d=38$, QAP obj=22]: the master played Rd4+; under μ our Rf7 plays the role of that game’s Rc4; cross-type roles: our Nd2 as their rh2, our Re7 as their Pc3; transferred move f7f8 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#3193 (ply 90) [structural distance $d=37$, QAP obj=22]: the master played Rd5+; under μ our Rf7 plays the role of that game’s Re5; cross-type roles: our Re7 as their Pe4, our Nd2 as their ra8; transferred move f7f8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rf8# (summed QAP objective 71, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.